Medical Data Analysis Pipeline Using Google Cloud BigQuery and SQL

Introduction:

The Medical Data Analysis Pipeline is a data analytics project developed using Google Cloud BigQuery and SQL. The project focuses on cleaning, transforming, analyzing, and validating medical data to generate meaningful insights.

The dataset contains information about patients, including age, gender, blood group, medical condition, admission type, hospital, billing amount, length of stay, and test results.

Google Cloud BigQuery is used to process and analyze the data using SQL queries. The cleaned data is then used to identify patient demographics, medical conditions, hospital performance, billing patterns, and length of hospital stays.

The final analysis helps transform raw medical data into meaningful business insights that can support healthcare reporting and decision-making.

Objectives:

The main objective of this project is to develop a medical data analysis pipeline using Google Cloud BigQuery and SQL.

The specific objectives are:

To clean and transform raw medical data.

To standardize gender values and remove unnecessary spaces from text fields.

To convert age, billing amount, and length of stay into appropriate data types.

To analyze the total number of patients and their demographic characteristics.

To identify the most common medical conditions and admission types.

To analyze patient distribution across hospitals and blood groups.

To calculate average, minimum, and maximum billing amounts.

To analyze average, minimum, and maximum hospital stay durations.

To validate the cleaned dataset by comparing row counts and checking column data types.

To generate final business insights that support healthcare data analysis and reporting.

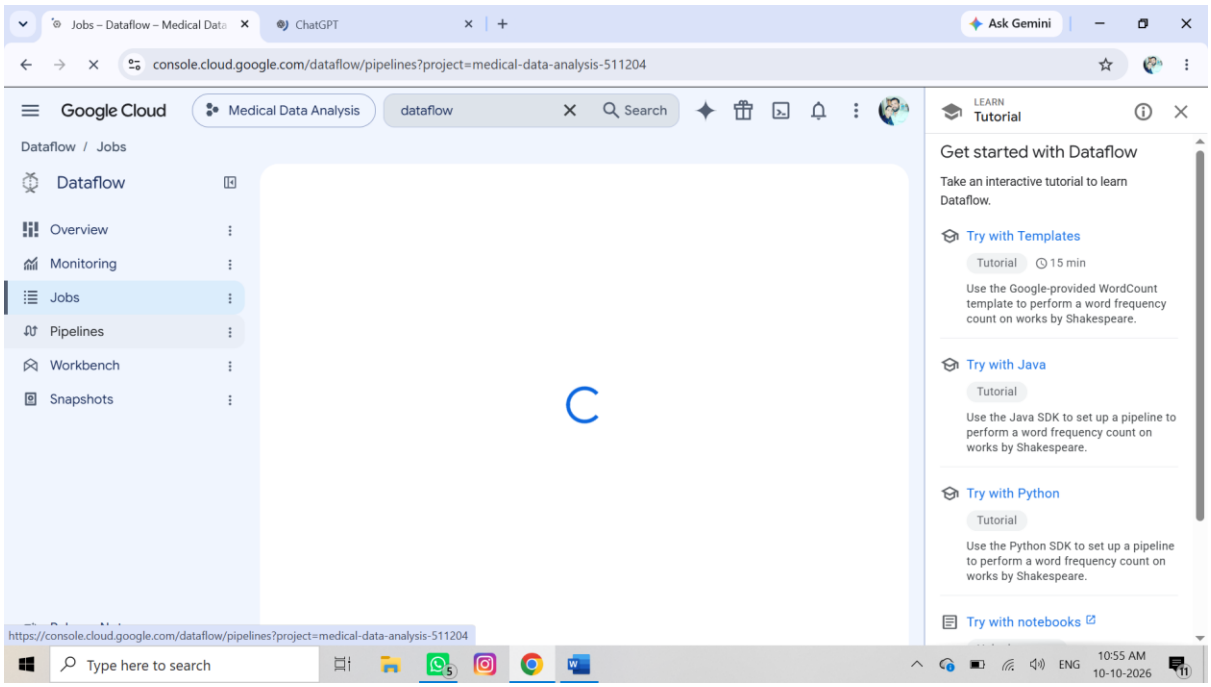
WorkFlow:

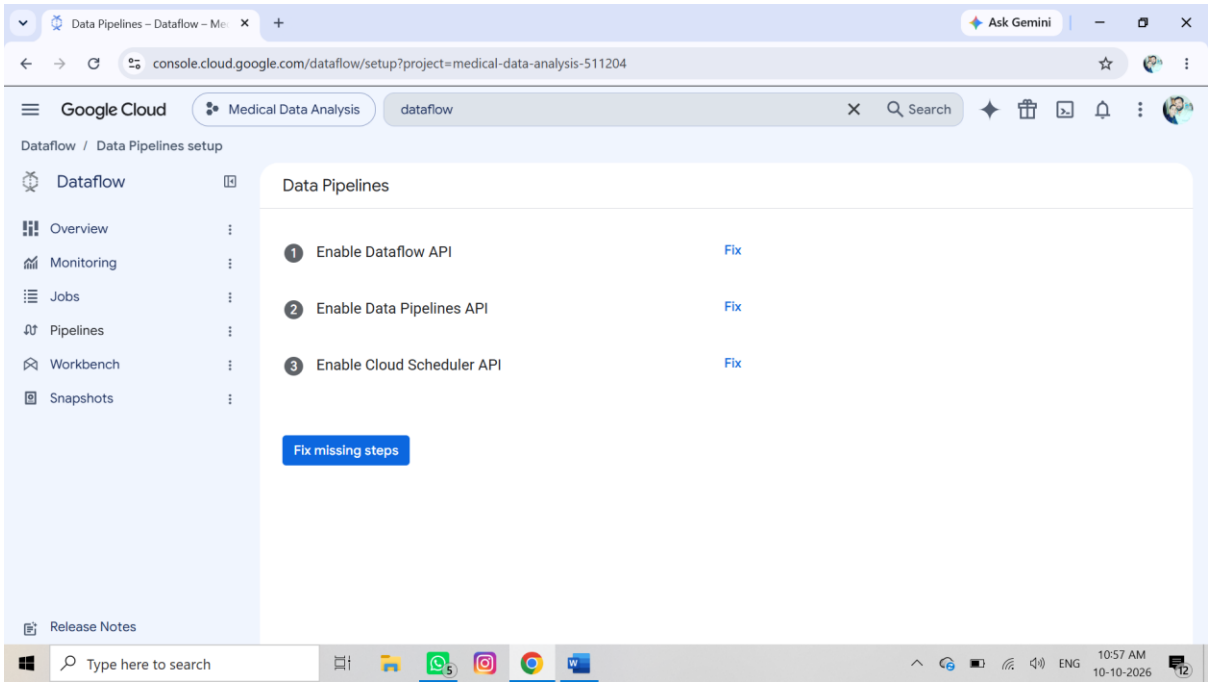
Workflow Definition

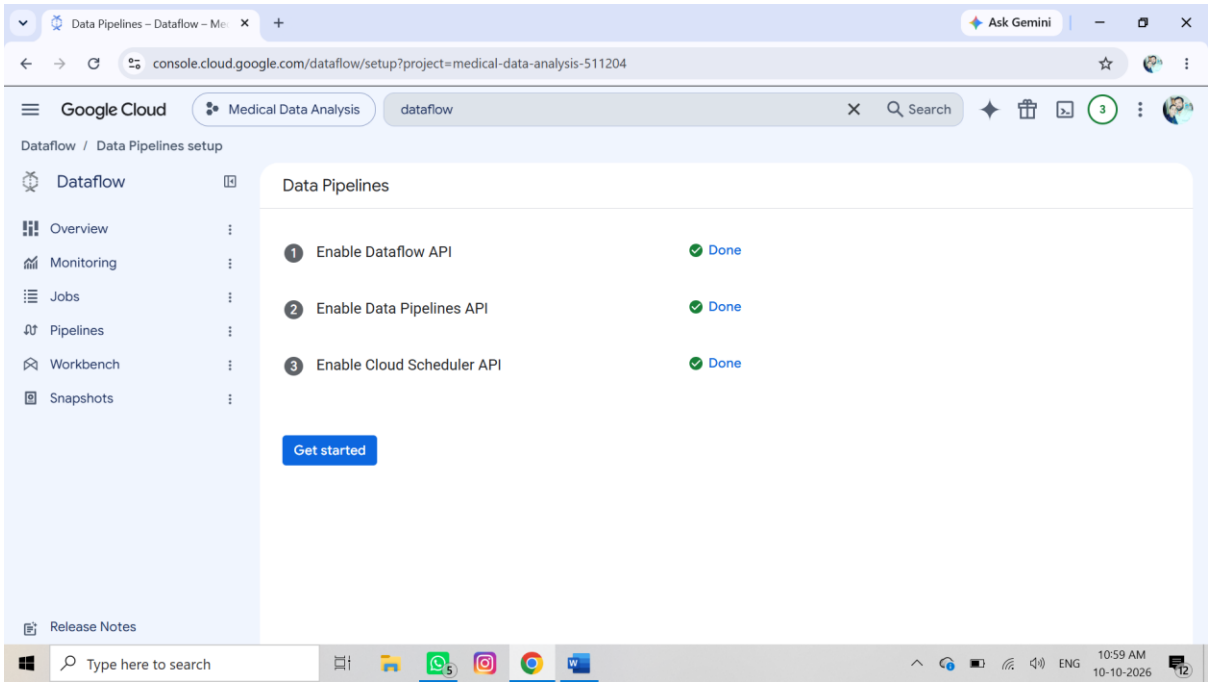
The Medical Data Analysis Workflow is a step-by-step process of collecting, cleaning, transforming, analyzing, and validating medical data using Google Cloud BigQuery and SQL. The cleaned data is used to generate meaningful healthcare insights and develop an interactive dashboard using Looker Studio.

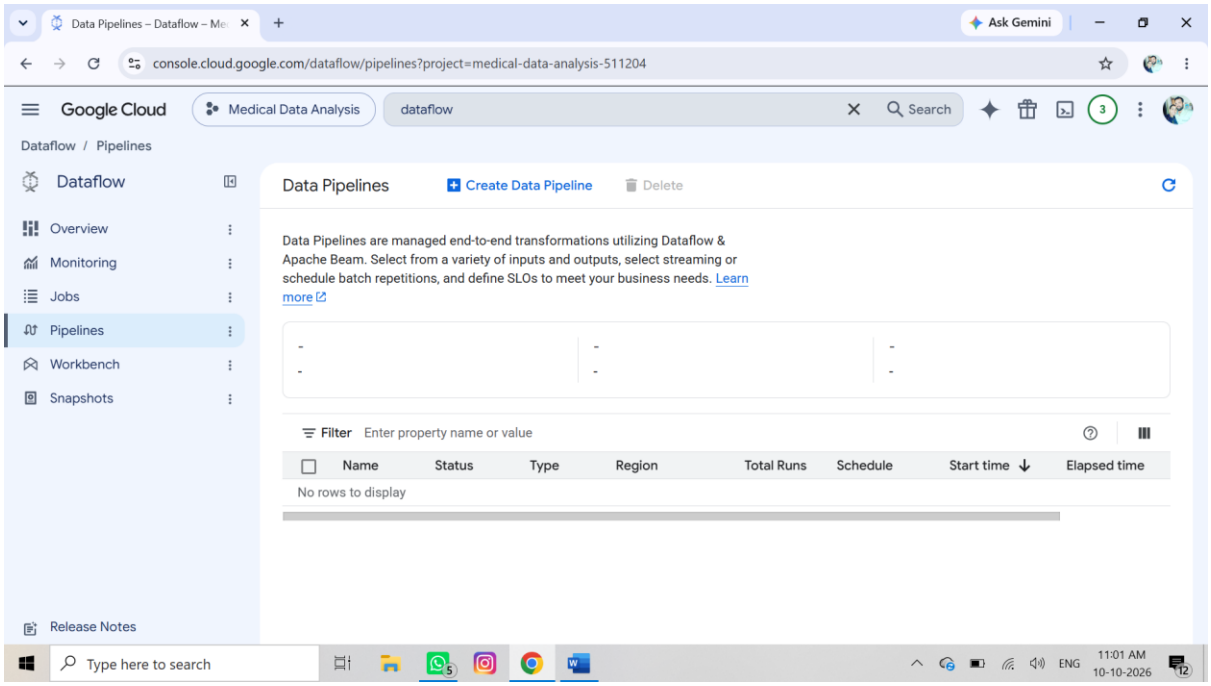
Workflow Steps:

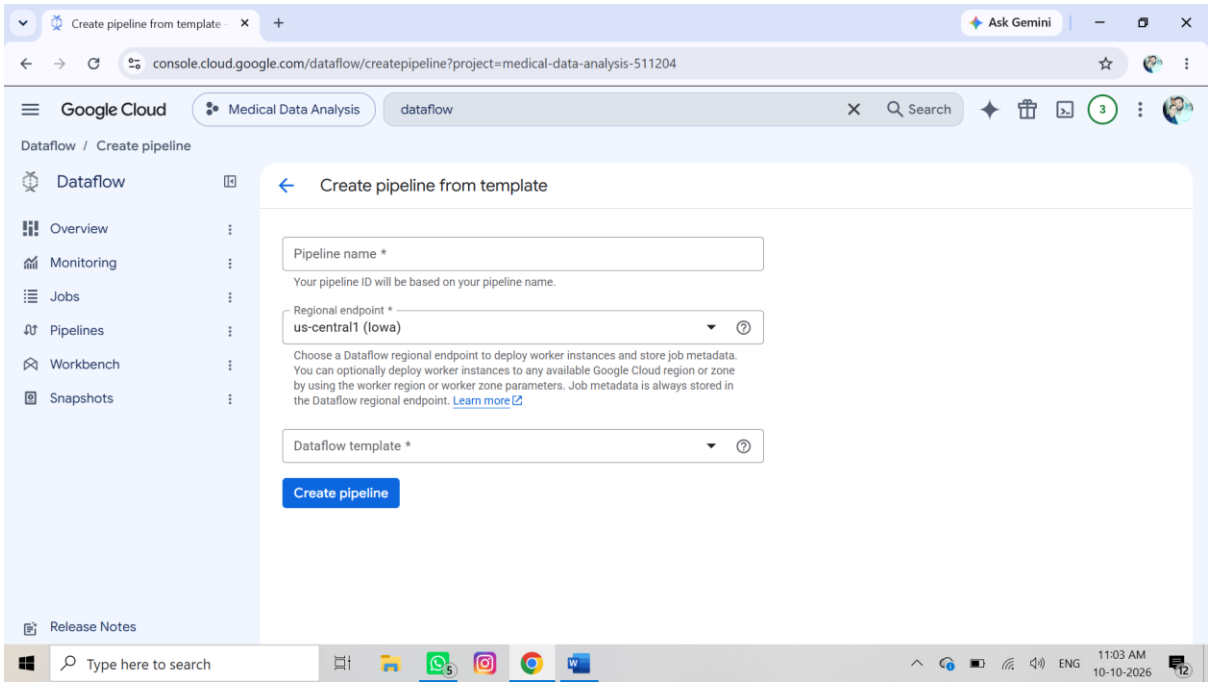
- Data Source: Import the raw medical dataset into BigQuery.
- Data Preparation: Inspect the dataset and identify data quality issues.
- Data Cleaning: Handle data types, standardize gender values, and remove unnecessary spaces.
- Data Transformation: Create the cleaned table named medical_data_cleaned.
- Data Analysis: Execute SQL queries to analyze patients, medical conditions, hospitals, and billing.
- Data Validation: Compare row counts and verify column data types.
- Dashboard Development: Visualize key metrics using Looker Studio.
- Final Insights: Summarize the findings to support data-driven healthcare decisions.

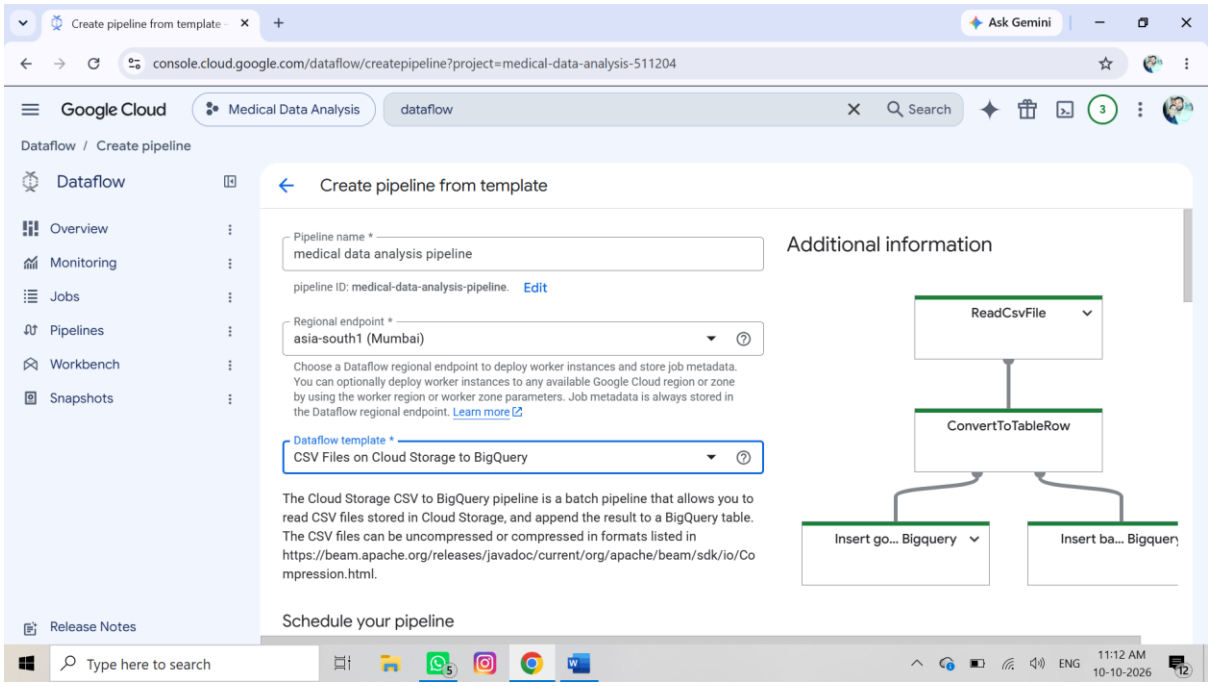


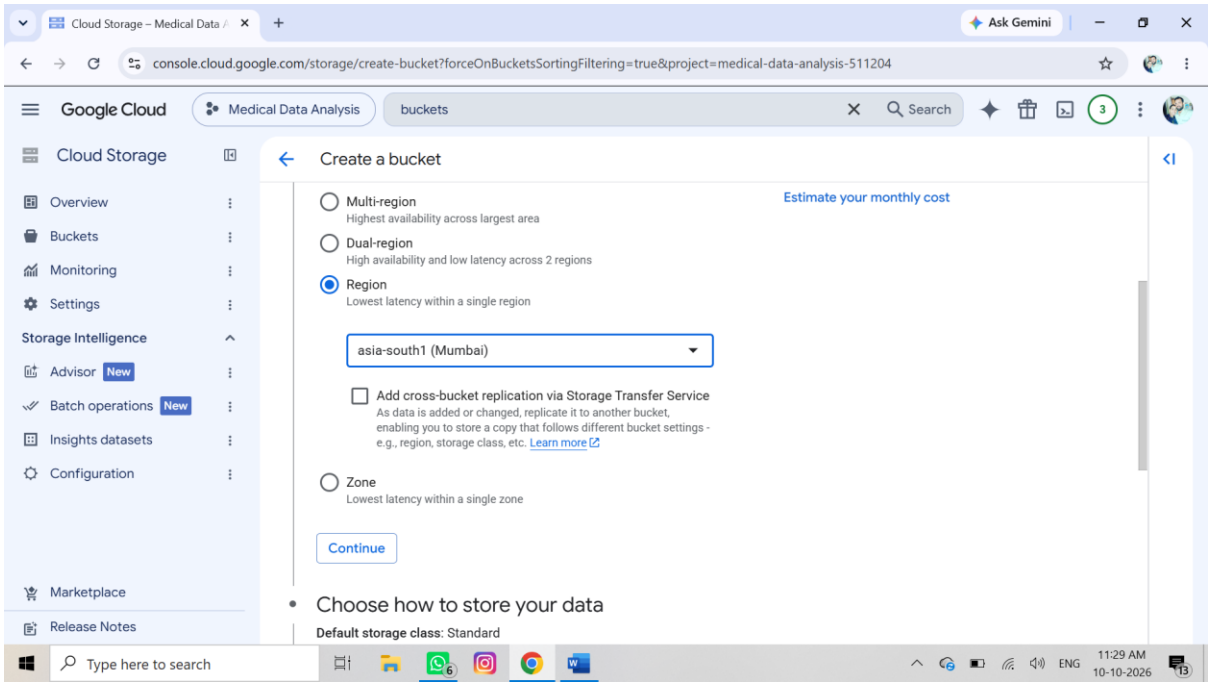

Data Pipelines – Dataflow – Me
Ask Gemini
console.cloud.google.com/dataflow/setup?project=medical-data-analysis-511204
Google Cloud
Medical Data Analysis
dataflow
Search
Dataflow / Data Pipelines setup
Dataflow
Overview
Monitoring
Jobs
Pipelines
Workbench
Snapshots
Release Notes
Data Pipelines
1 Enable Dataflow API
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2 Enable Data Pipelines API
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3 Enable Cloud Scheduler API
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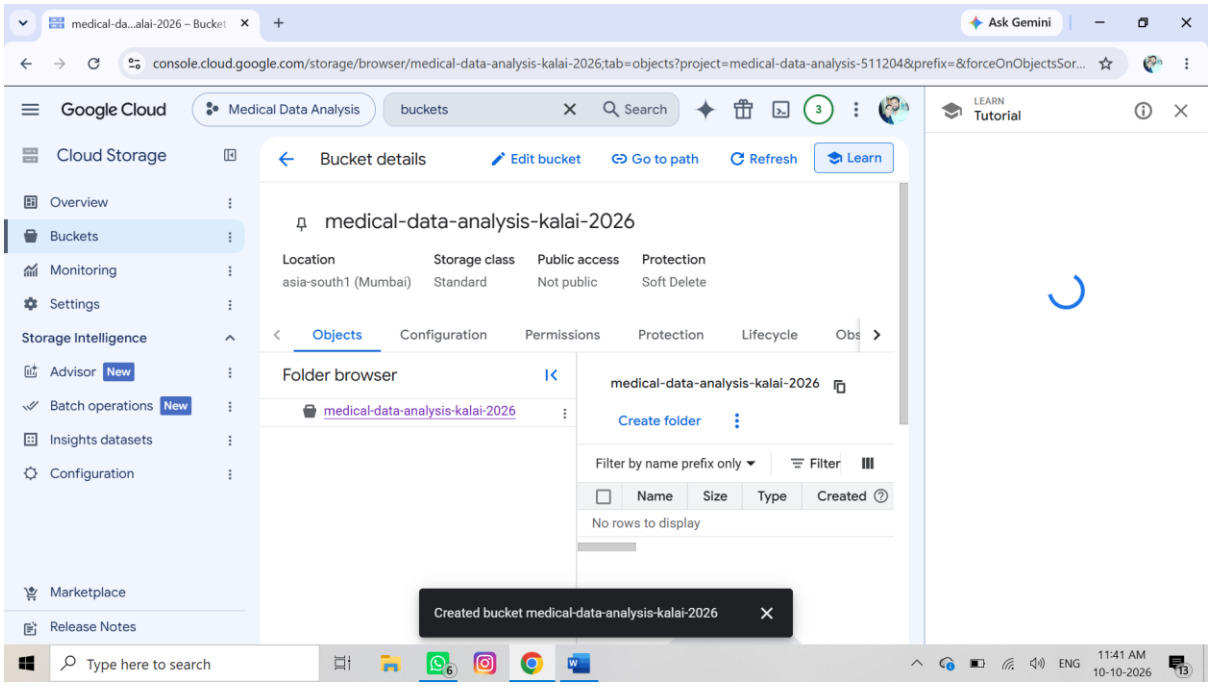

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Dataflow / Data Pipelines setup
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Release Notes
Data Pipelines
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Done
2 Enable Data Pipelines API
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3 Enable Cloud Scheduler API
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Data Pipelines
Create Data Pipeline
Delete
Data Pipelines are managed end-to-end transformations utilizing Dataflow & Apache Beam. Select from a variety of inputs and outputs, select streaming or schedule batch repetitions, and define SLOs to meet your business needs. Learn more
Filter
Enter property name or value
Name
Status
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Region
Total Runs
Schedule
Start time
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Create pipeline from template
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Dataflow
Overview
Monitoring
Jobs
Pipelines
Workbench
Snapshots
Release Notes
Create pipeline from template
Pipeline name *
Your pipeline ID will be based on your pipeline name.
Regional endpoint *
us-central1 (Iowa)
Choose a Dataflow regional endpoint to deploy worker instances and store job metadata. You can optionally deploy worker instances to any available Google Cloud region or zone by using the worker region or worker zone parameters. Job metadata is always stored in the Dataflow regional endpoint. Learn more
Dataflow template *
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Create pipeline from template
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Dataflow
Overview
Monitoring
Jobs
Pipelines
Workbench
Snapshots
Release Notes
Create pipeline from template
Pipeline name *
medical data analysis pipeline
pipeline ID: medical-data-analysis-pipeline. Edit
Regional endpoint *
asia-south1 (Mumbai)
Choose a Dataflow regional endpoint to deploy worker instances and store job metadata. You can optionally deploy worker instances to any available Google Cloud region or zone by using the worker region or worker zone parameters. Job metadata is always stored in the Dataflow regional endpoint. Learn more
Dataflow template *
CSV Files on Cloud Storage to BigQuery
The Cloud Storage CSV to BigQuery pipeline is a batch pipeline that allows you to read CSV files stored in Cloud Storage, and append the result to a BigQuery table. The CSV files can be uncompressed or compressed in formats listed in https://beam.apache.org/releases/javadoc/current/org/apache/beam/sdk/io/Compression.html.
Additional information
ReadCsvFile
ConvertToTableRow
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Schedule your pipeline
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Cloud Storage – Medical Data
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Google Cloud
Medical Data Analysis
buckets
Search
Cloud Storage
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Advisor New
Batch operations New
Insights datasets
Configuration
Marketplace
Release Notes
Create a bucket
Estimate your monthly cost
Multi-region
Highest availability across largest area
Dual-region
High availability and low latency across 2 regions
Region
Lowest latency within a single region
asia-south1 (Mumbai)
Add cross-bucket replication via Storage Transfer Service
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. Learn more
Zone
Lowest latency within a single zone
Continue
Choose how to store your data
Default storage class: Standard
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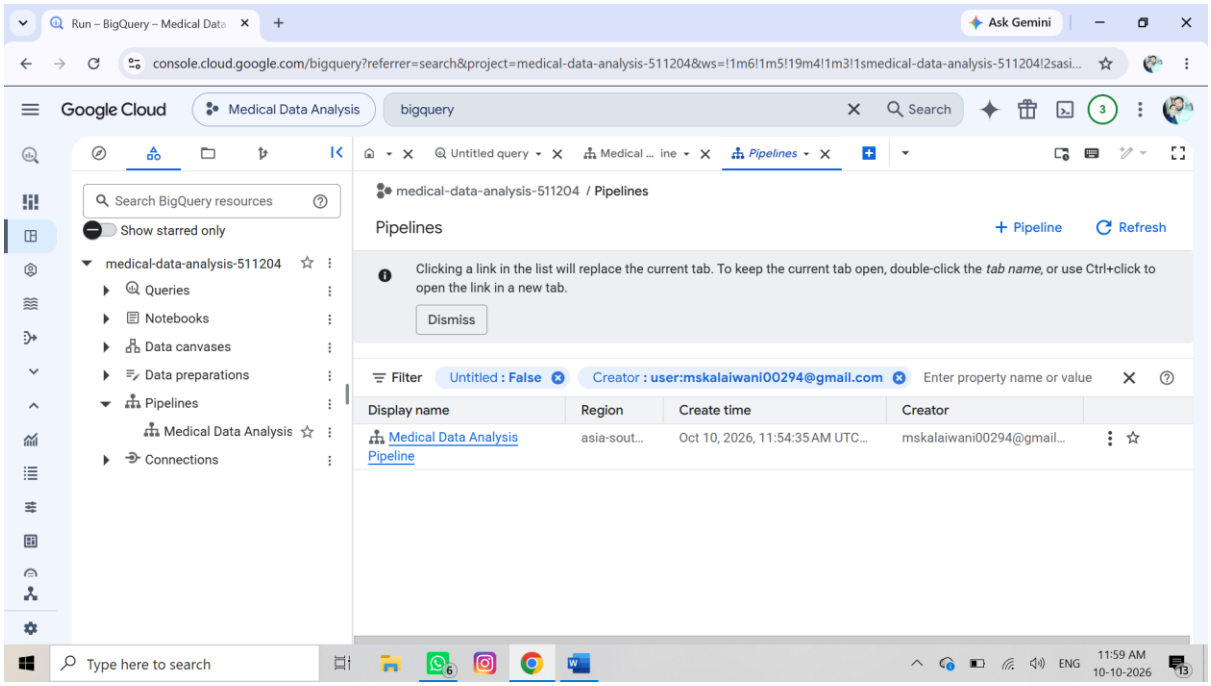


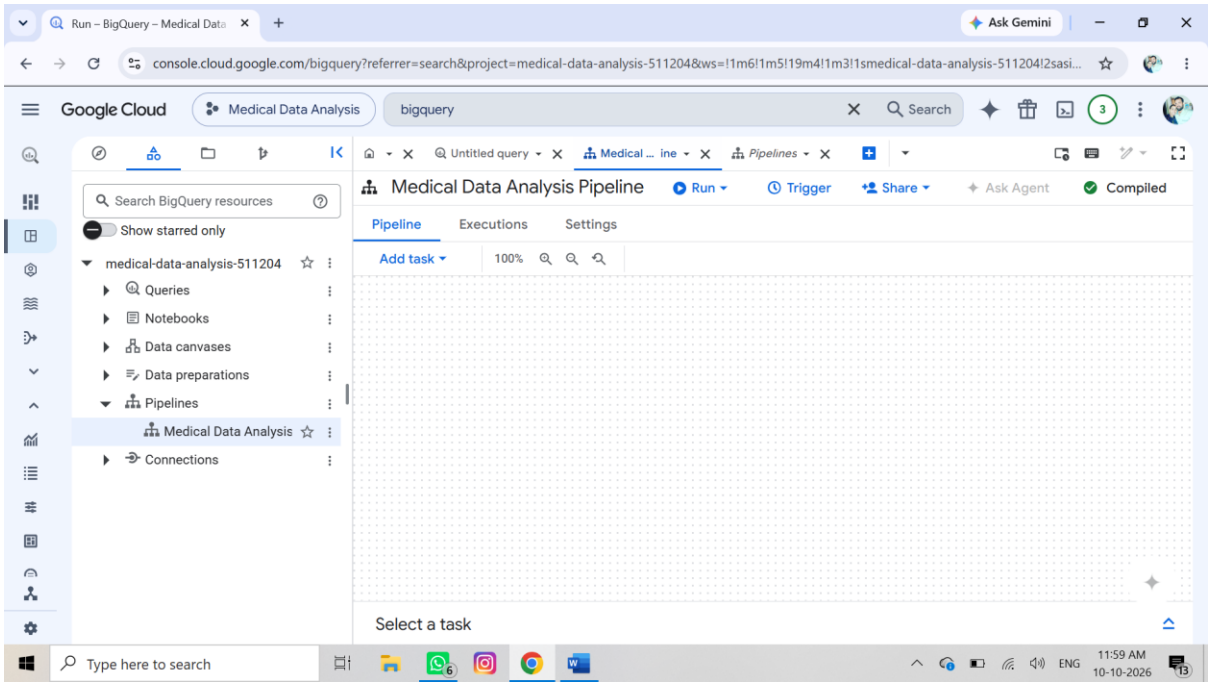
Pipeline Definition

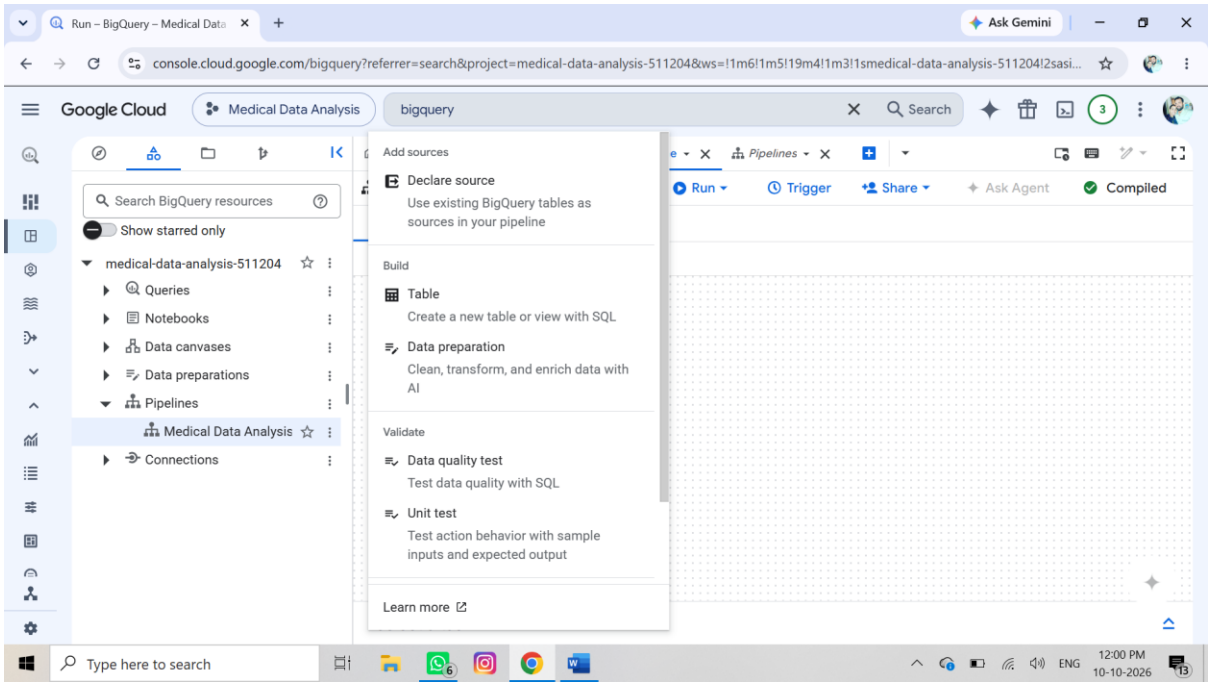
A Medical Data Analysis Pipeline is a sequence of steps used to prepare, clean, transform, analyze, and validate medical data using Google Cloud BigQuery and SQL. The pipeline converts raw medical data into structured information and meaningful insights for healthcare reporting and dashboard visualization.

Pipeline Steps:

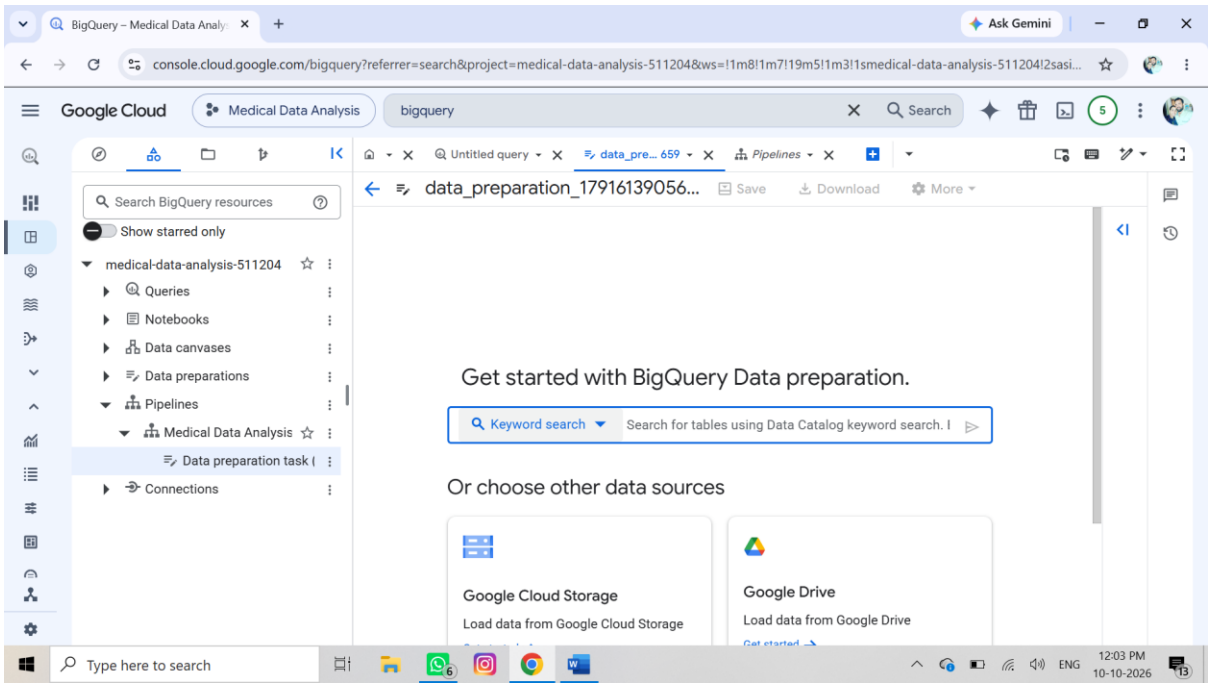
- Data Ingestion
- Data Preparation
- Data Cleaning and Transformation
- Data Storage in BigQuery
- SQL Data Analysis
- Data Validation
- Dashboard Creation using Looker Studio
- Final Business Insights

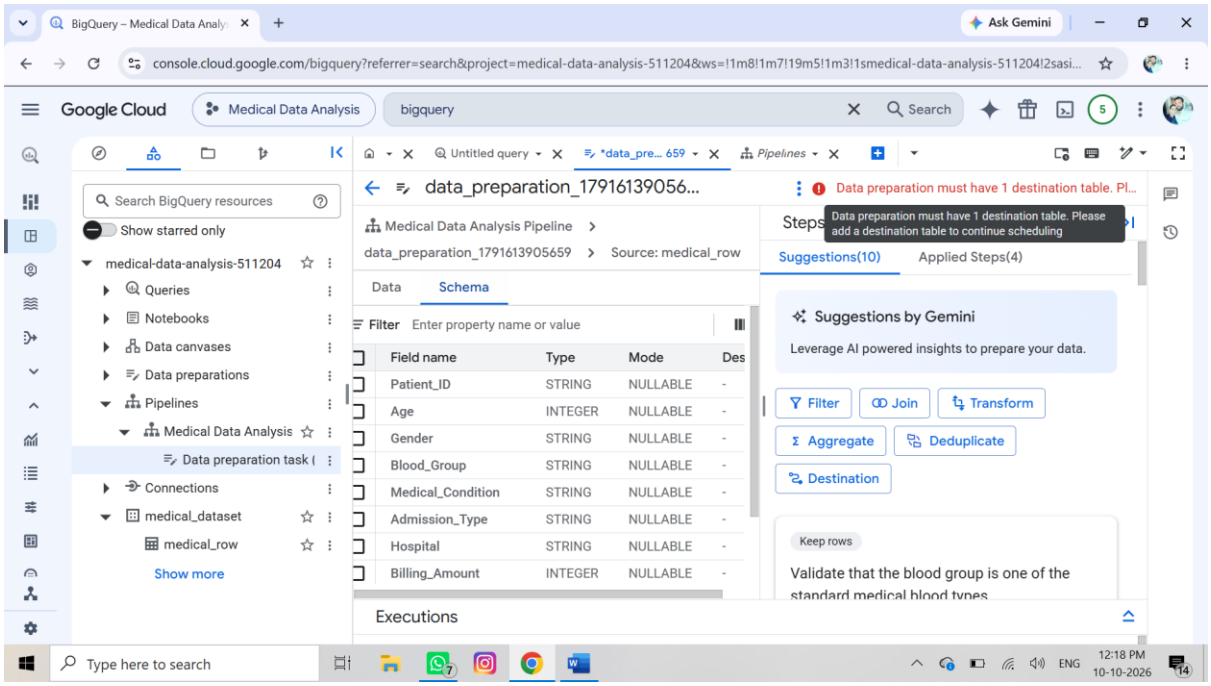

Run – BigQuery – Medical Data
Ask Gemini
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Display name
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Run – BigQuery – Medical Data
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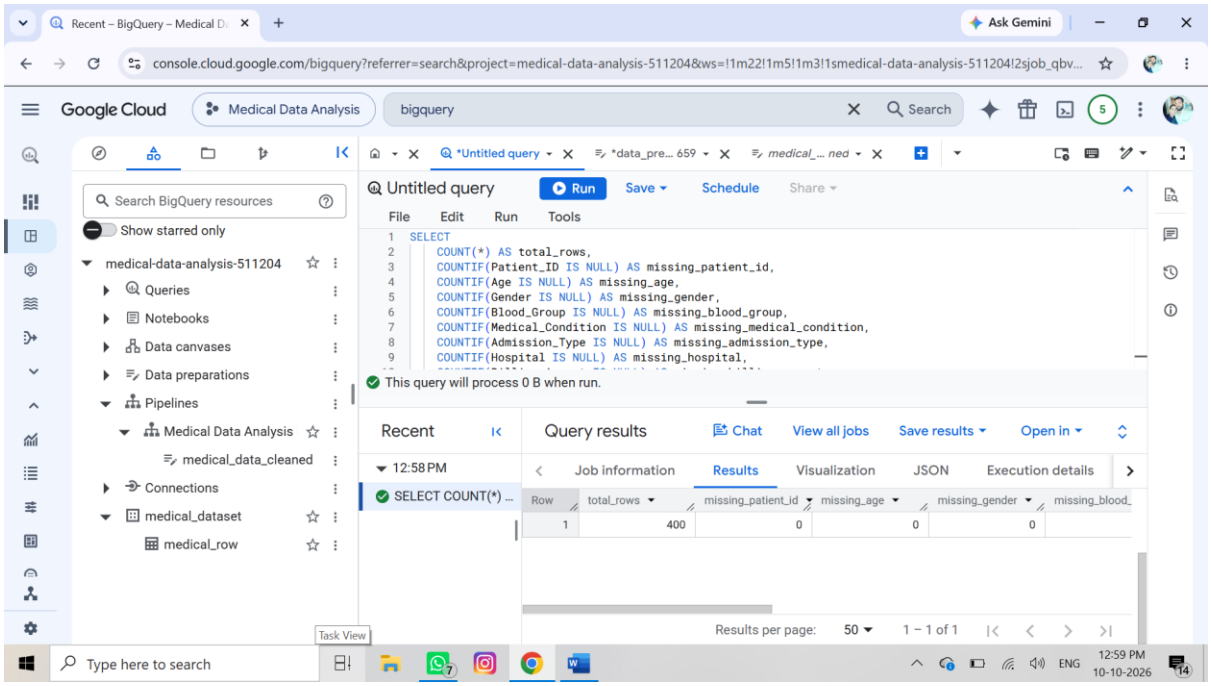


Data Preparation Step 1:

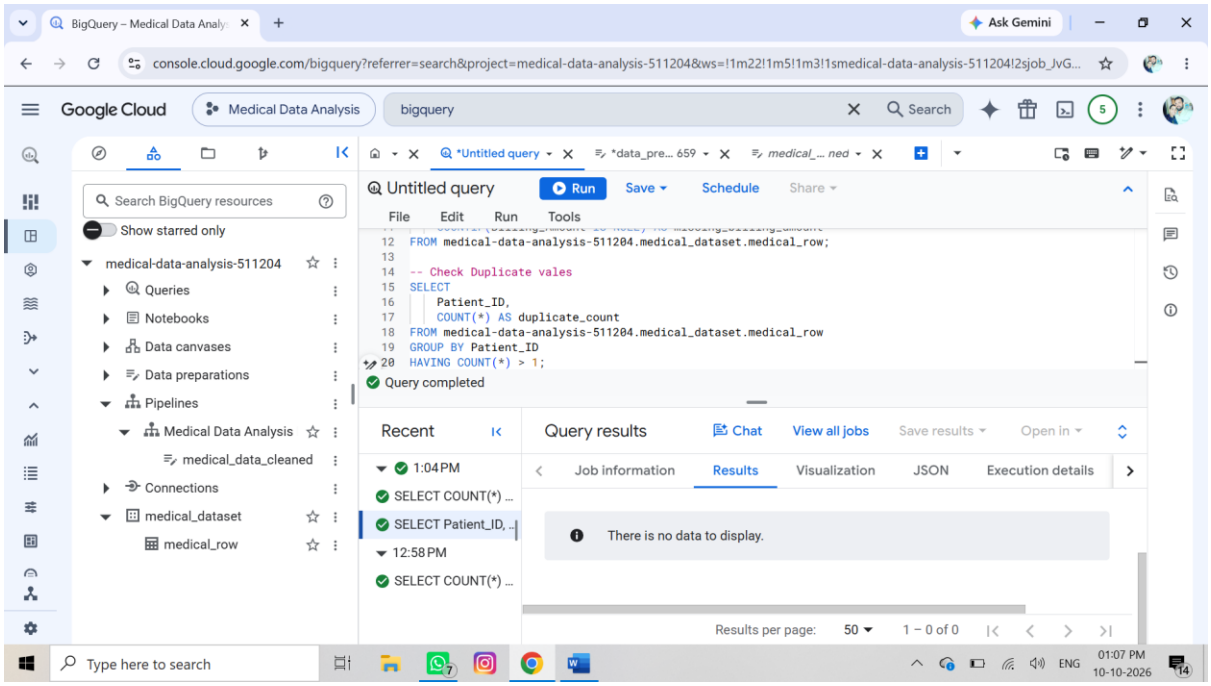


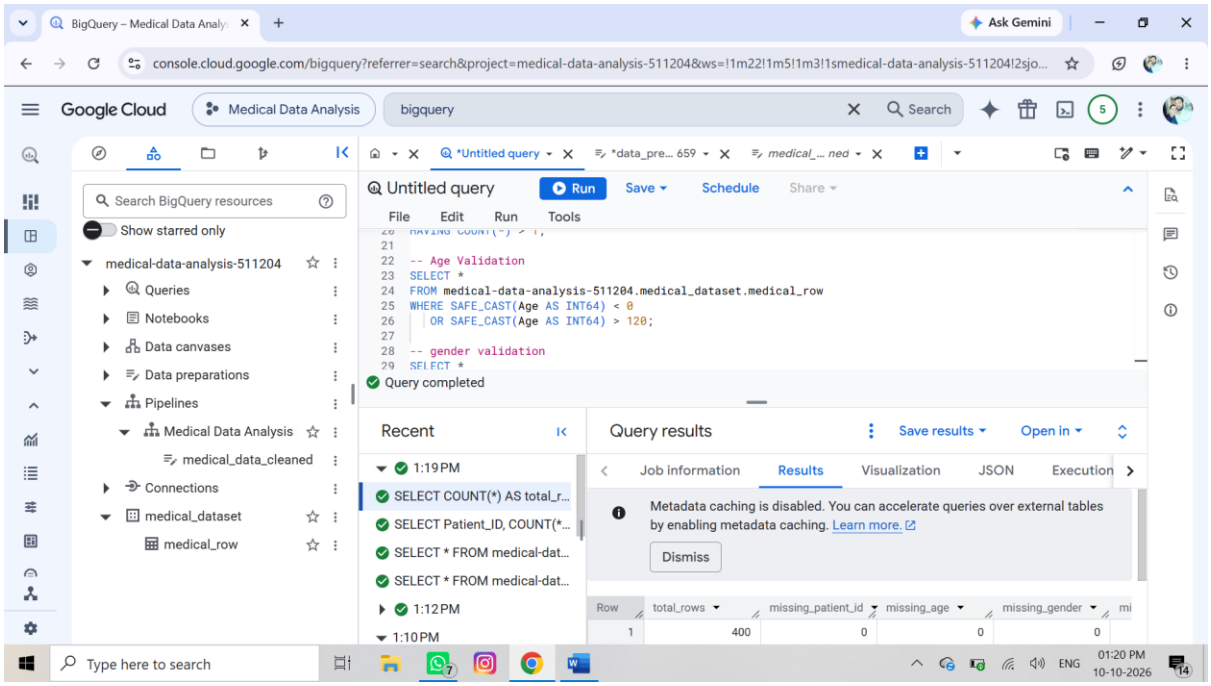


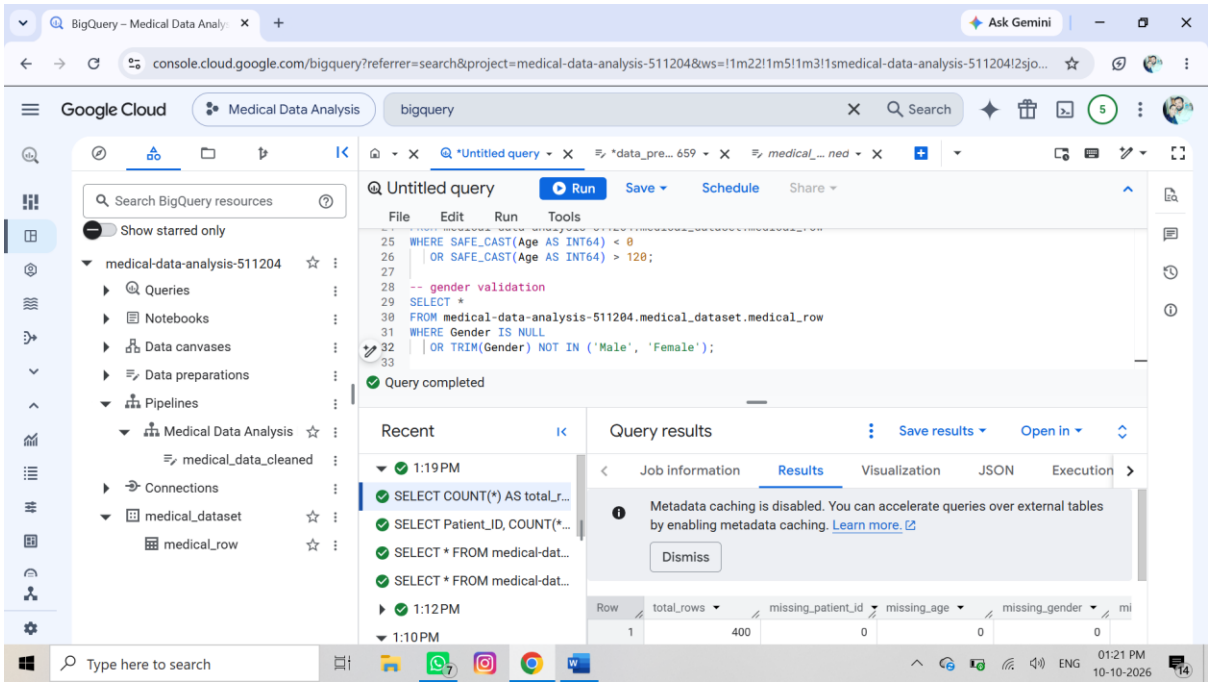
Check Missing Values

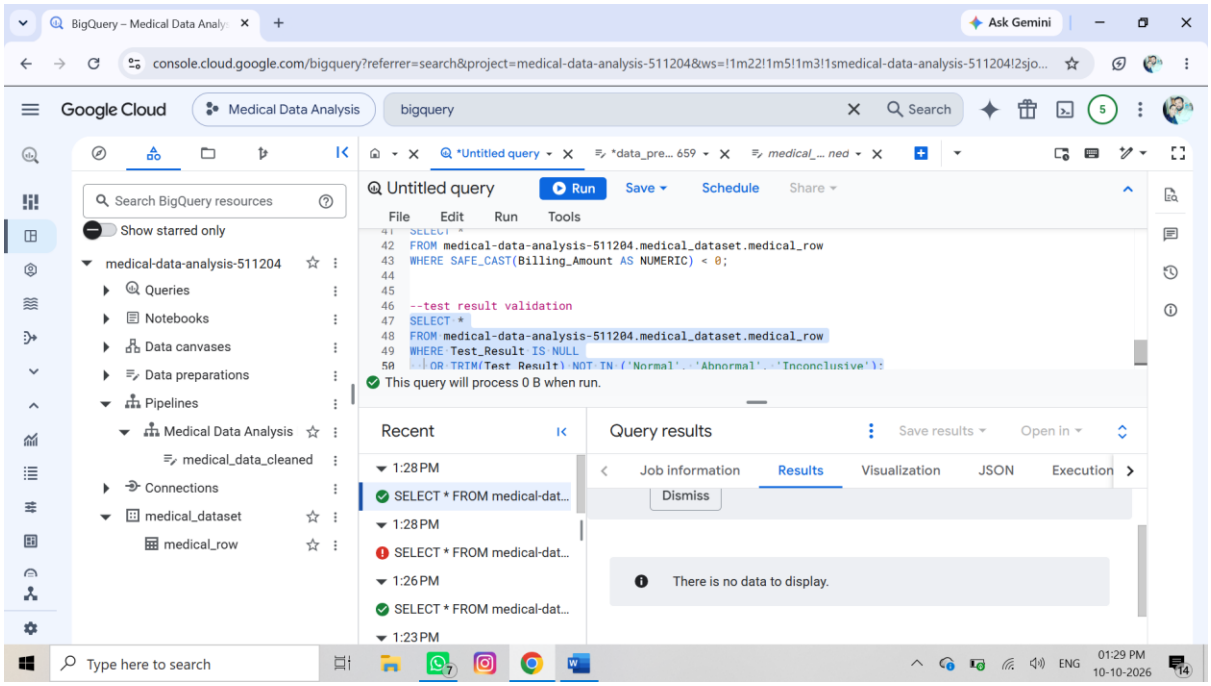


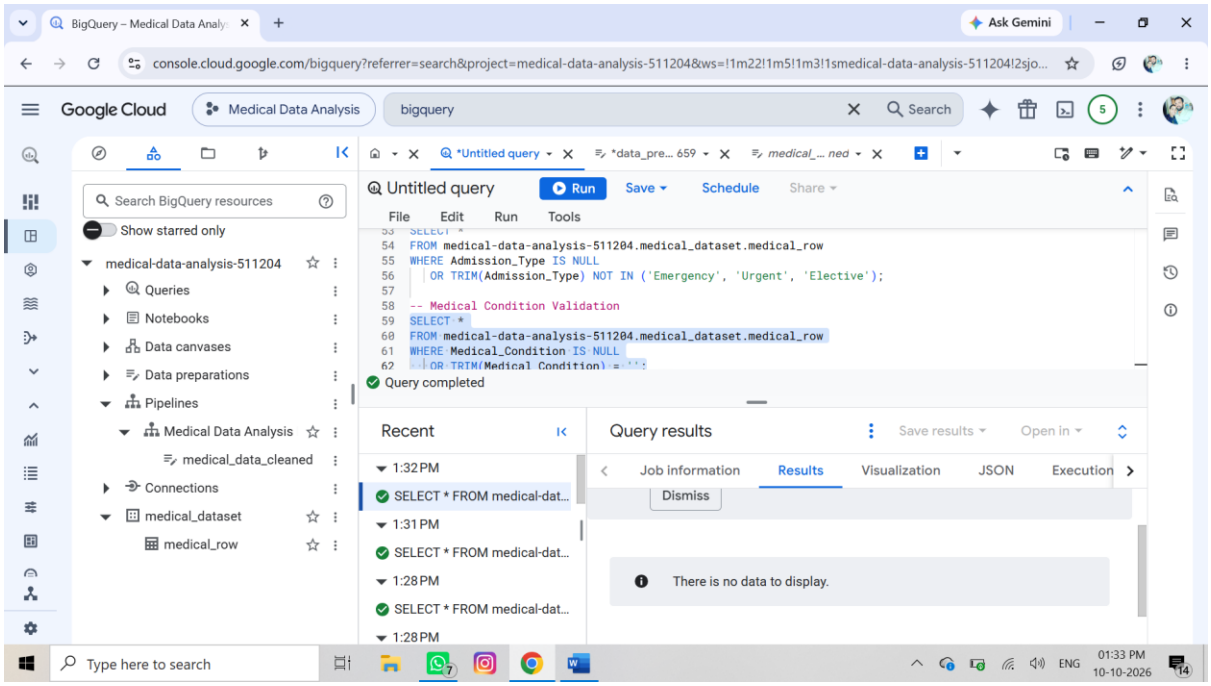
Check Duplicate records

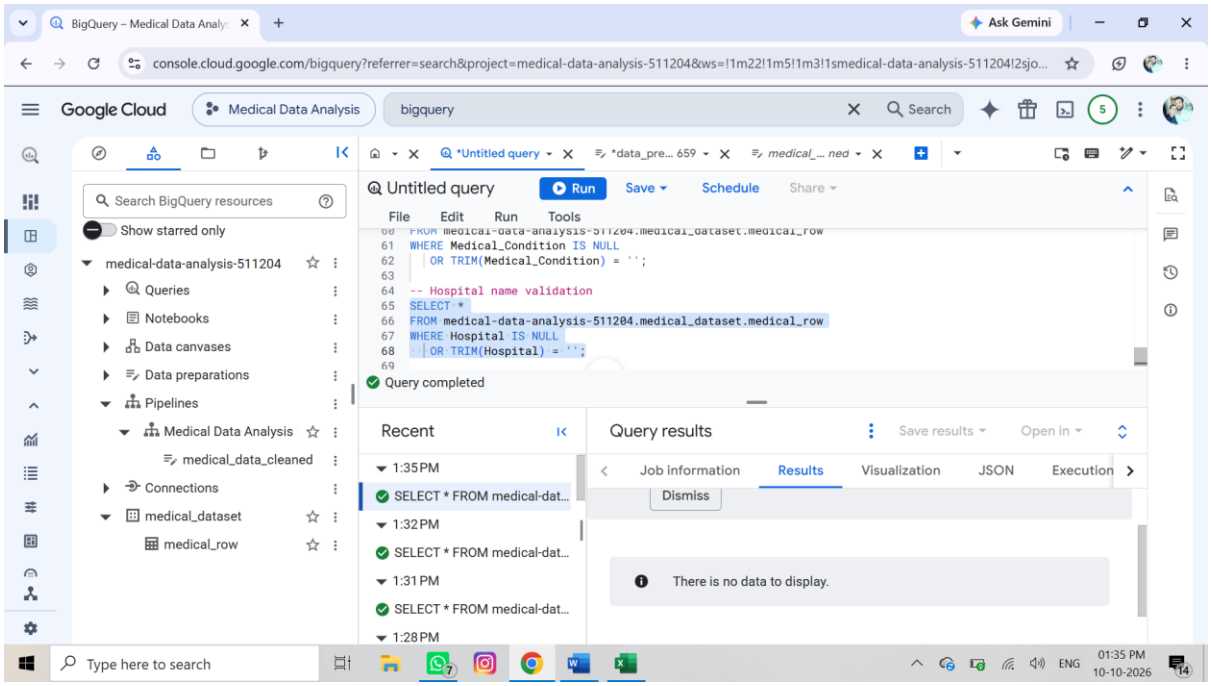


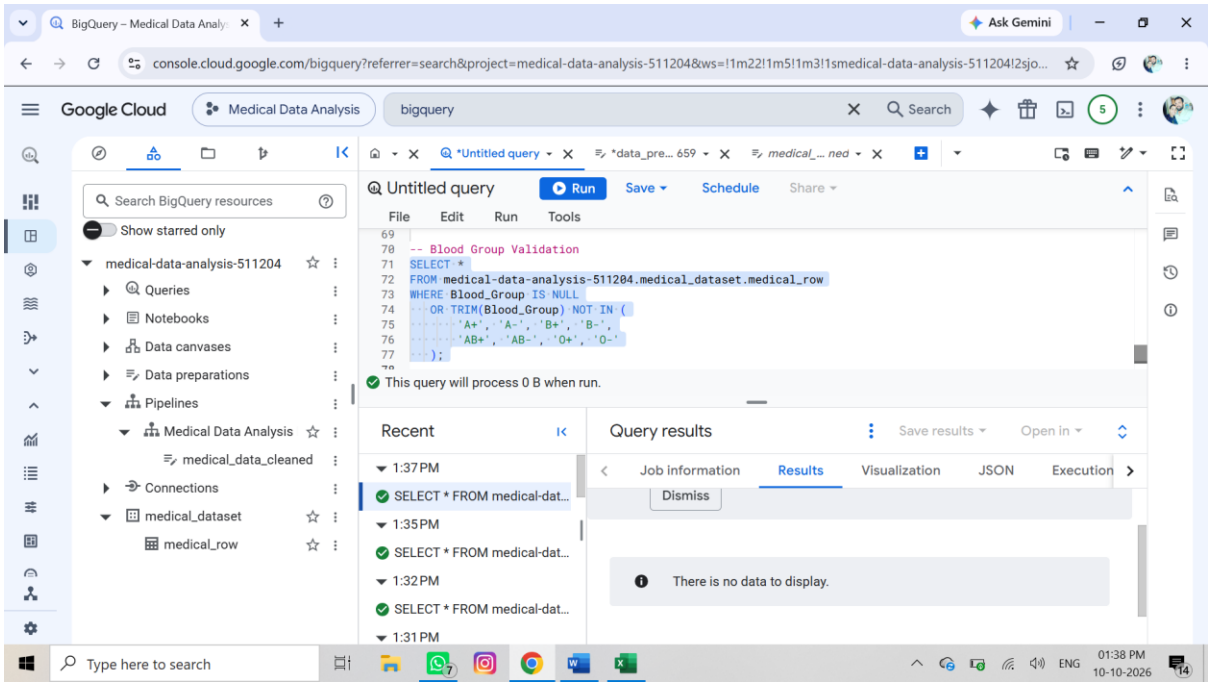

BigQuery – Medical Data Analy
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Medical Data Analysis
medical_data_cleaned
Connections
medical_dataset
medical_row
Untitled query
Run
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Schedule
Share
File
Edit
Run
Tools
-- Age Validation
SELECT *
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE SAFE_CAST(Age AS INT64) < 0
OR SAFE_CAST(Age AS INT64) > 120;
-- gender validation
SELECT *
Query completed
Recent
Query results
Save results
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1:19 PM
SELECT COUNT(*) AS total_r...
SELECT Patient_ID, COUNT(*...
SELECT * FROM medical-dat...
SELECT * FROM medical-dat...
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Job information
Results
Visualization
JSON
Execution
Metadata caching is disabled. You can accelerate queries over external tables by enabling metadata caching. Learn more.
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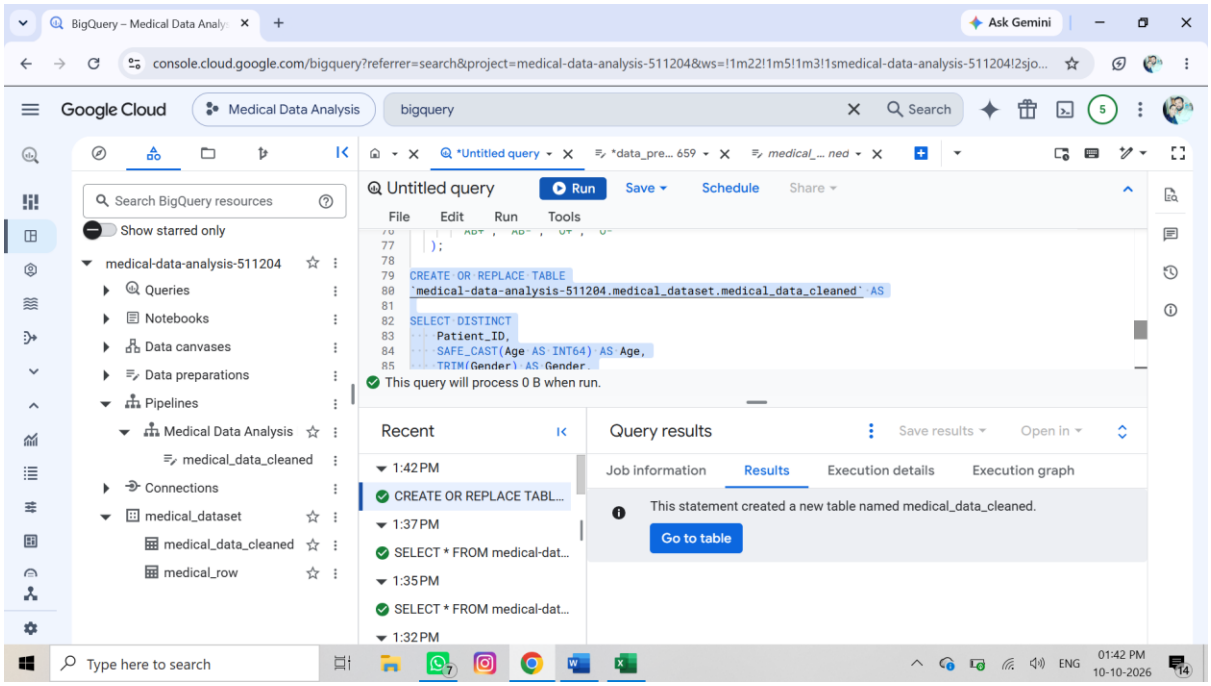

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WHERE SAFE_CAST(Age AS INT64) < 0
OR SAFE_CAST(Age AS INT64) > 120;
-- gender validation
SELECT *
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE Gender IS NULL
OR TRIM(Gender) NOT IN ('Male', 'Female');
Query completed
Recent
Query results
Save results
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SELECT COUNT(*) AS total_r...
SELECT Patient_ID, COUNT(*...
SELECT * FROM medical-dat...
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BigQuery – Medical Data Analy
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Tools
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE SAFE_CAST(Billing_Amount AS NUMERIC) < 0;
--test result validation
SELECT *
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE Test_Result IS NULL
OR TRIM(Test_Result) NOT IN ('Normal','Abnormal','Inconclusive');
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SELECT * FROM medical-dat...
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BigQuery – Medical Data Analy
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Run
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Tools
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE Admission_Type IS NULL
OR TRIM(Admission_Type) NOT IN ('Emergency', 'Urgent', 'Elective');
-- Medical Condition Validation
SELECT *
FROM medical-data-analysis-511204.medical_dataset.medical_row
WHERE Medical_Condition IS NULL
OR TRIM(Medical_Condition) = '';
Query completed
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SELECT * FROM medical-dat...
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SELECT * FROM medical-dat...
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SELECT * FROM medical-dat...
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BigQuery – Medical Data Analy
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Pipelines
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Run
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Schedule
Share
File
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Run
Tools
61 WHERE Medical_Condition IS NULL
62 OR TRIM(Medical_Condition) = '';
63
64 -- Hospital name validation
65 SELECT *
66 FROM medical-data-analysis-511204.medical_dataset.medical_row
67 WHERE Hospital IS NULL
68 OR TRIM(Hospital) = '';
Query completed
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SELECT * FROM medical-dat...
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SELECT * FROM medical-dat...
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BigQuery – Medical Data Analy
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Connections
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medical_row
Untitled query
Run
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Tools
70 -- Blood Group Validation
71 SELECT *
72 FROM medical-data-analysis-511204.medical_dataset.medical_row
73 WHERE Blood_Group IS NULL
74 OR TRIM(Blood_Group) NOT IN (
75 'A+', 'A-', 'B+', 'B-',
76 'AB+', 'AB-', 'O+', 'O-'
77);
This query will process 0 B when run.
Recent
1:37 PM
SELECT * FROM medical-dat...
1:35 PM
SELECT * FROM medical-dat...
1:32 PM
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Job information
Results
Visualization
JSON
Execution
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Analysis SQL

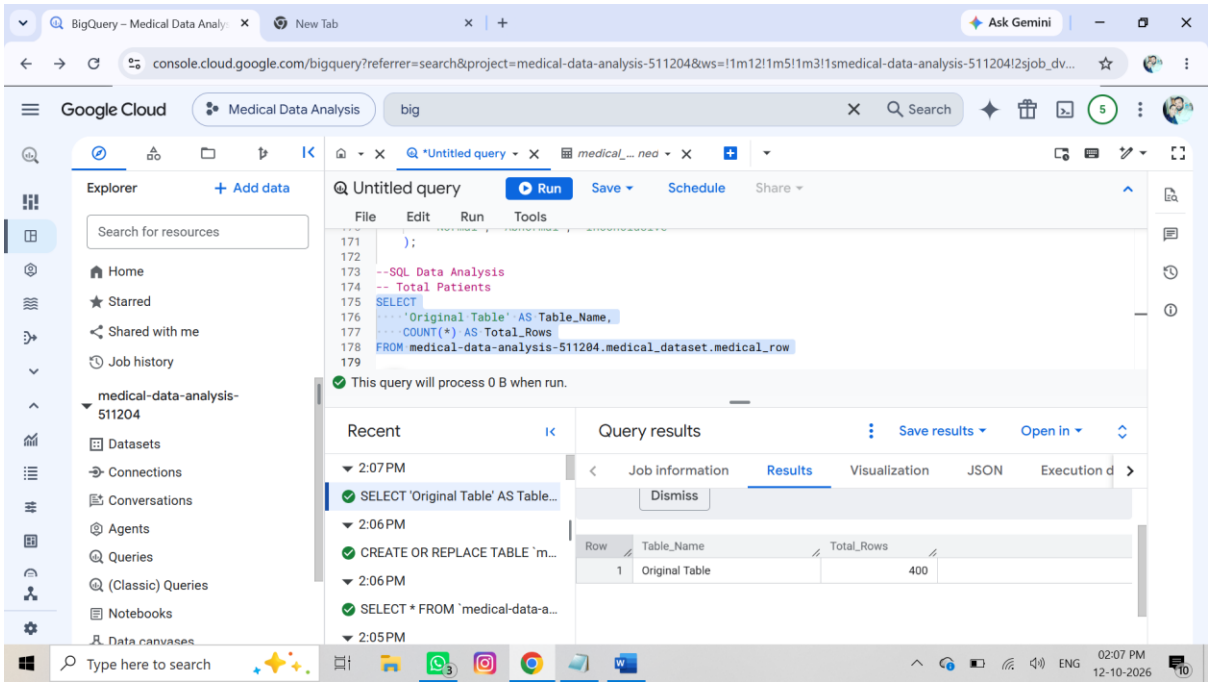
SQL Data Query

Total Patients

SELECT

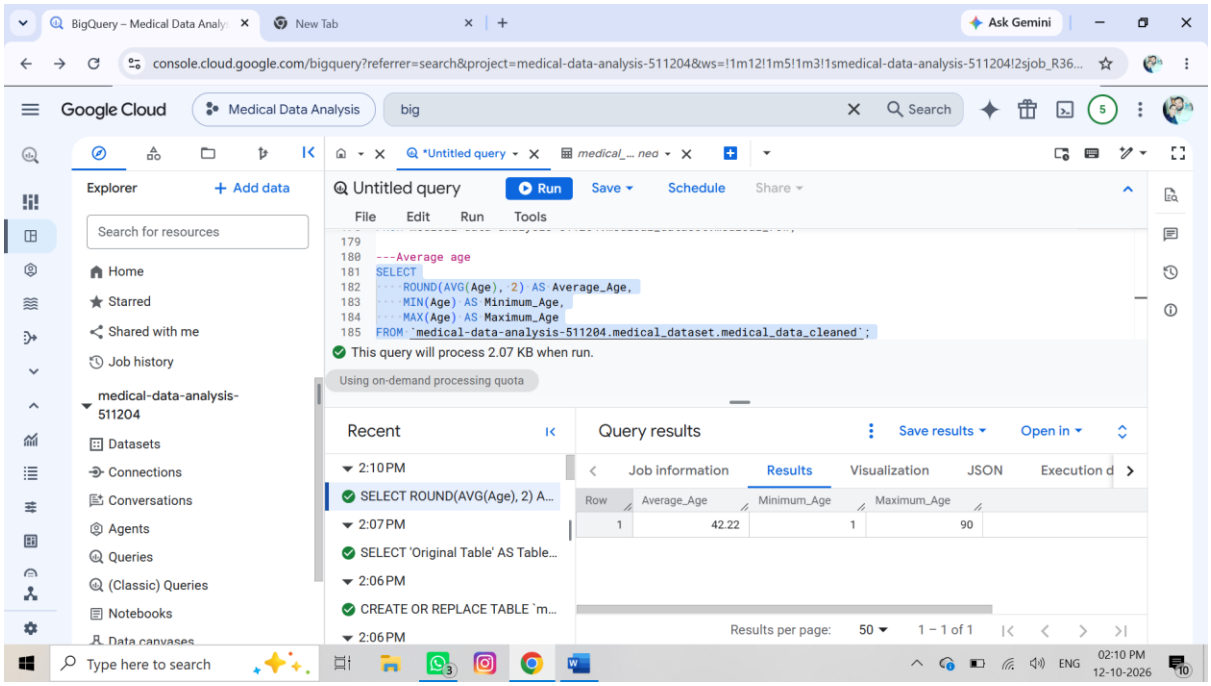
COUNT(DISTINCT Patient_ID) AS Total_Patients

FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`;



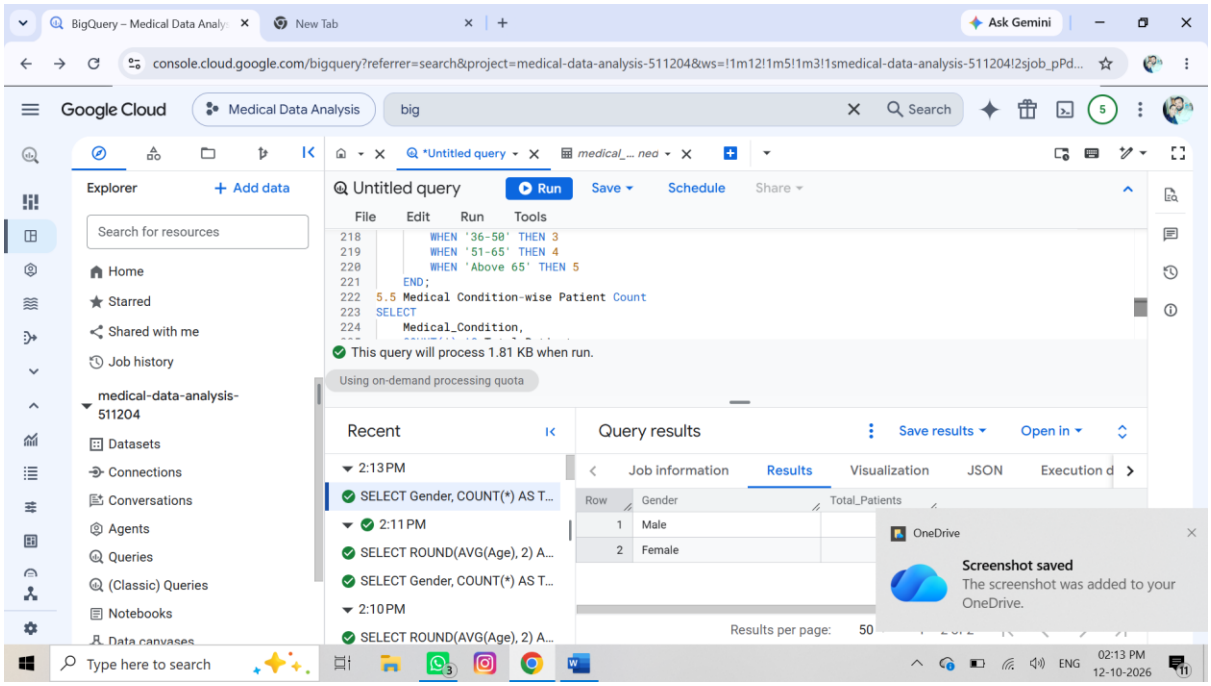
Average Age

```
SELECT
    ROUND(AVG(Age), 2) AS Average_Age,
    MIN(Age) AS Minimum_Age,
    MAX(Age) AS Maximum_Age
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`;
```



Gender-wise Patient Count

```
SELECT
    Gender,
    COUNT(*) AS Total_Patients
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`
GROUP BY Gender
ORDER BY Total_Patients DESC;
```



Age-wise Patient Analysis

```
SELECT
    CASE
        WHEN Age < 18 THEN 'Under 18'
        WHEN Age BETWEEN 18 AND 35 THEN '18-35'
        WHEN Age BETWEEN 36 AND 50 THEN '36-50'
        WHEN Age BETWEEN 51 AND 65 THEN '51-65'
        ELSE 'Above 65'
    END AS Age_Group,

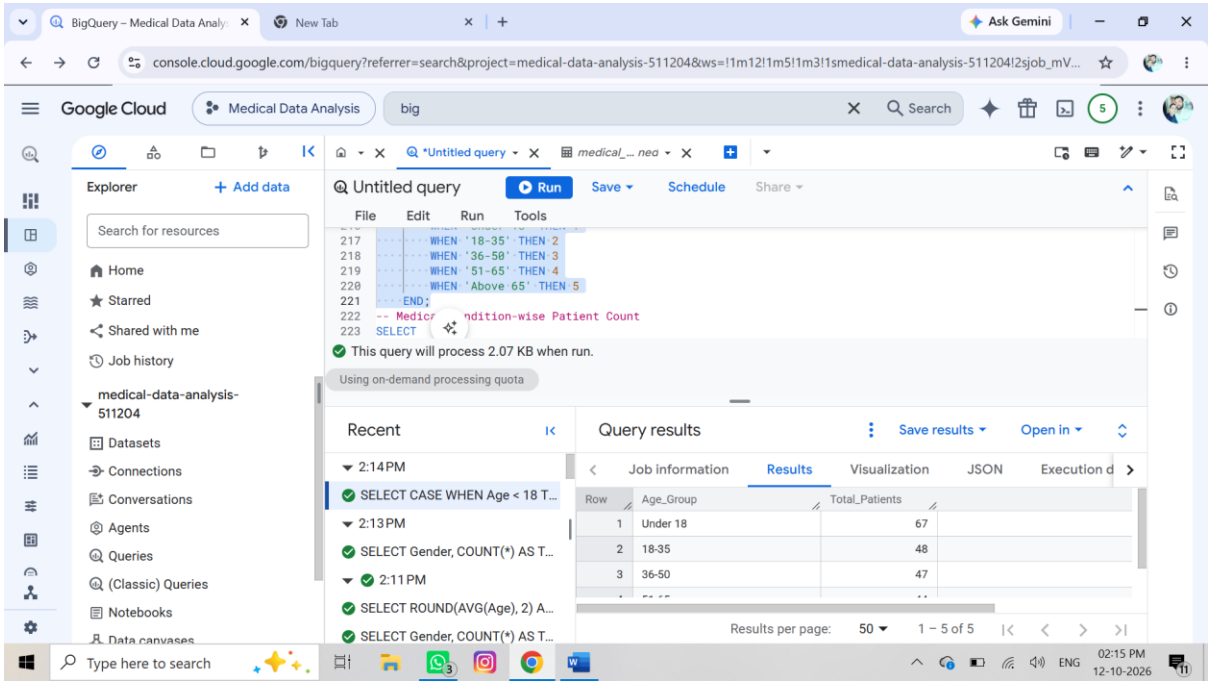
    COUNT(*) AS Total_Patients

FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`

GROUP BY Age_Group

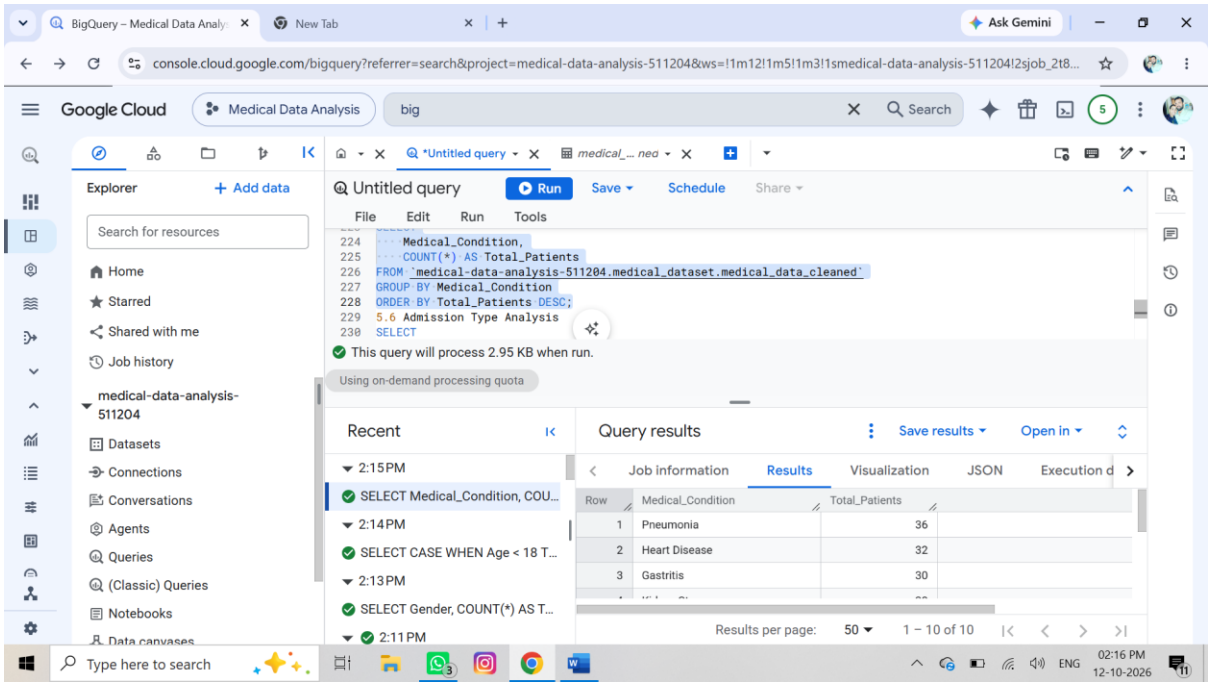
ORDER BY
    CASE Age_Group
        WHEN 'Under 18' THEN 1
```

```
    WHEN '18-35' THEN 2
    WHEN '36-50' THEN 3
    WHEN '51-65' THEN 4
    WHEN 'Above 65' THEN 5
  END;
```



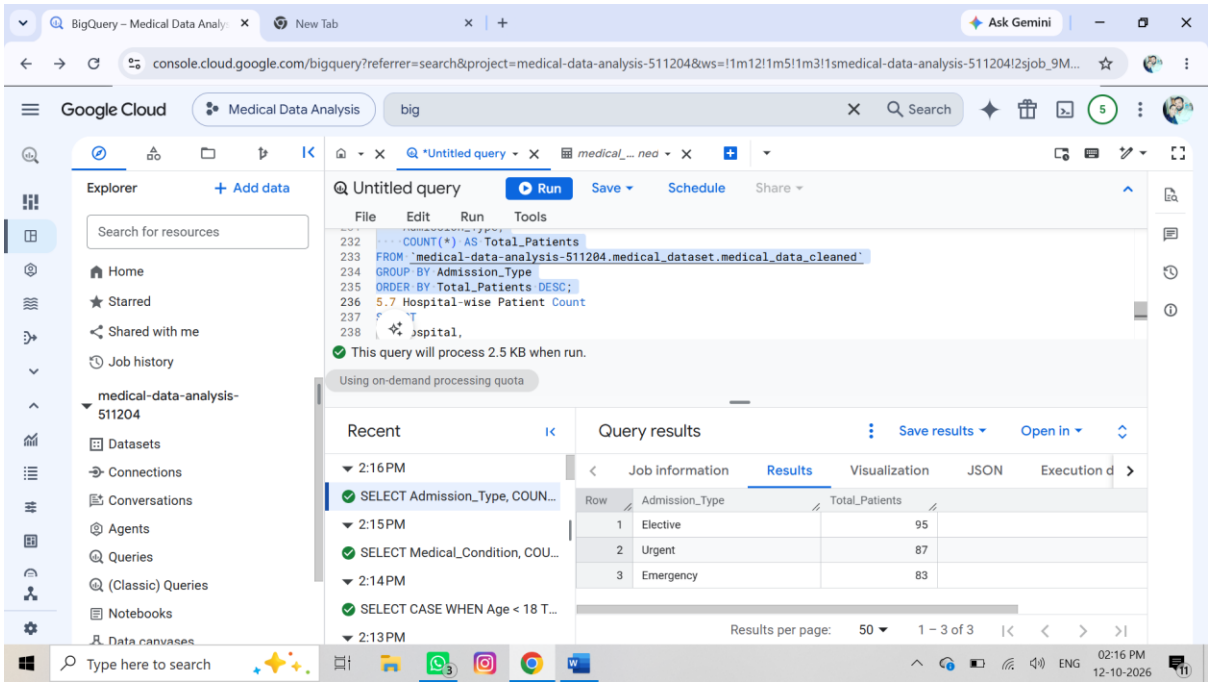
Medical Condition-wise Patient Count

```
SELECT
    Medical_Condition,
    COUNT(*) AS Total_Patients
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`
GROUP BY Medical_Condition
ORDER BY Total_Patients DESC;
```



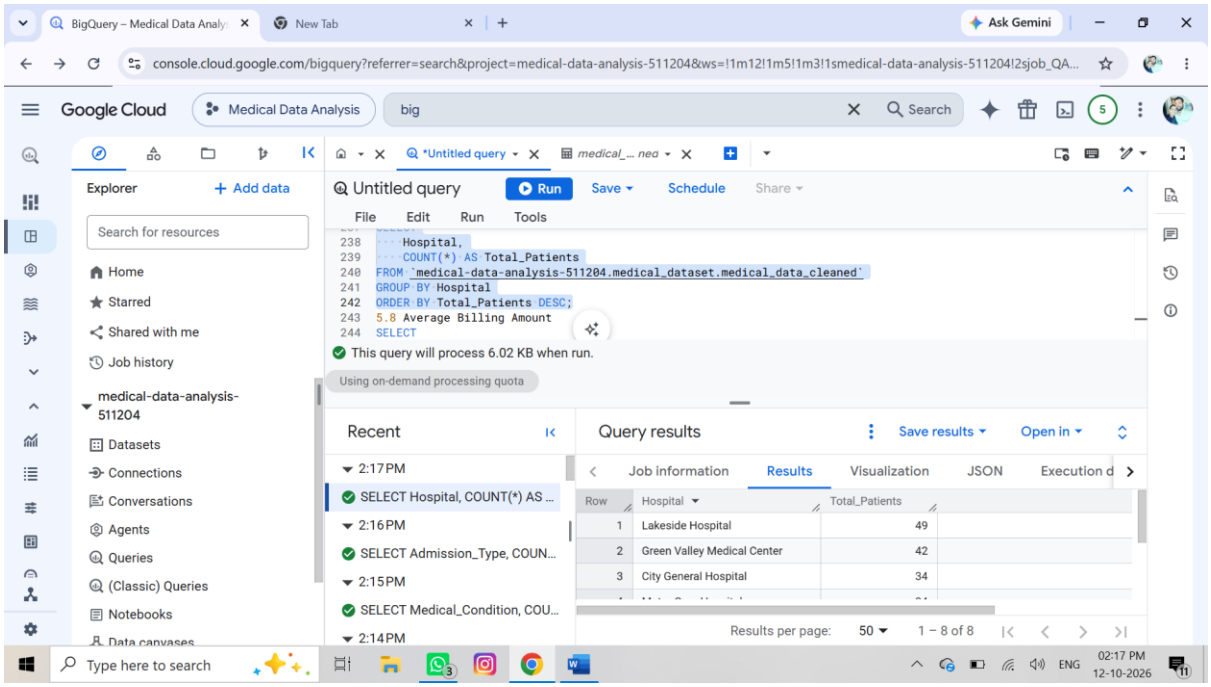
Admission Type Analysis

```
SELECT
    Admission_Type,
    COUNT(*) AS Total_Patients
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`
GROUP BY Admission_Type
ORDER BY Total_Patients DESC;
```



Hospital-wise Patient Count

```
SELECT
    Hospital,
    COUNT(*) AS Total_Patients
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`
GROUP BY Hospital
ORDER BY Total_Patients DESC;
```

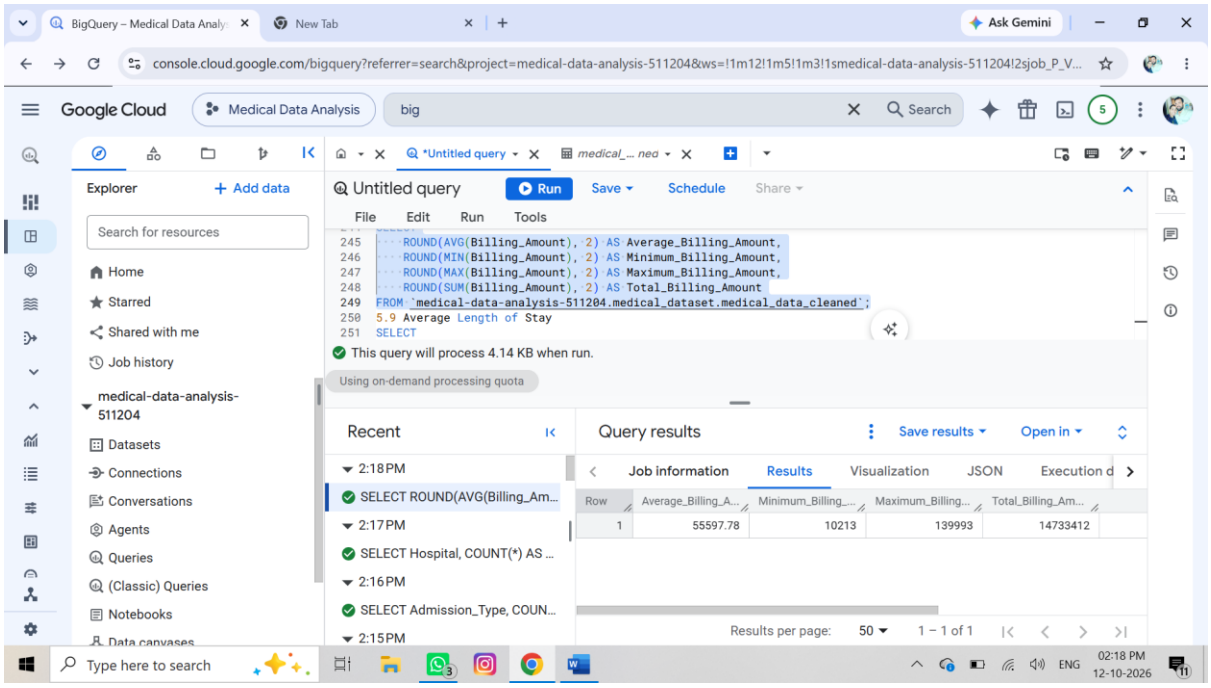


Average Billing Amount

```
SELECT
    ROUND(AVG(Billing_Amount), 2) AS Average_Billing_Amount,
    ROUND(MIN(Billing_Amount), 2) AS Minimum_Billing_Amount,
    ROUND(MAX(Billing_Amount), 2) AS Maximum_Billing_Amount,
    ROUND(SUM(Billing_Amount), 2) AS Total_Billing_Amount
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`;
```

Average Length of Stay

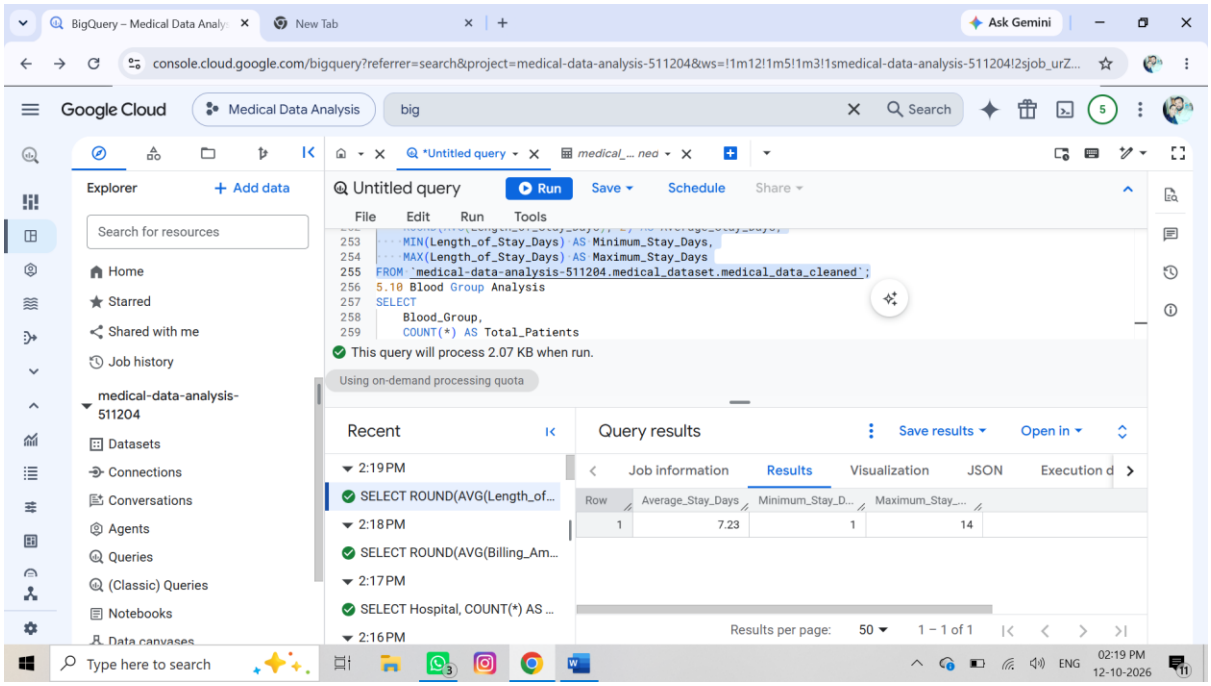
```
SELECT
    ROUND(AVG(Length_of_Stay_Days), 2) AS Average_Stay_Days,
    MIN(Length_of_Stay_Days) AS Minimum_Stay_Days,
    MAX(Length_of_Stay_Days) AS Maximum_Stay_Days
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`;
```



Blood Group Analysis

```
SELECT
    Blood_Group,
    COUNT(*) AS Total_Patients
FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`
GROUP BY Blood_Group
```

ORDER BY Total_Patients DESC;



Test Result Analysis

SELECT

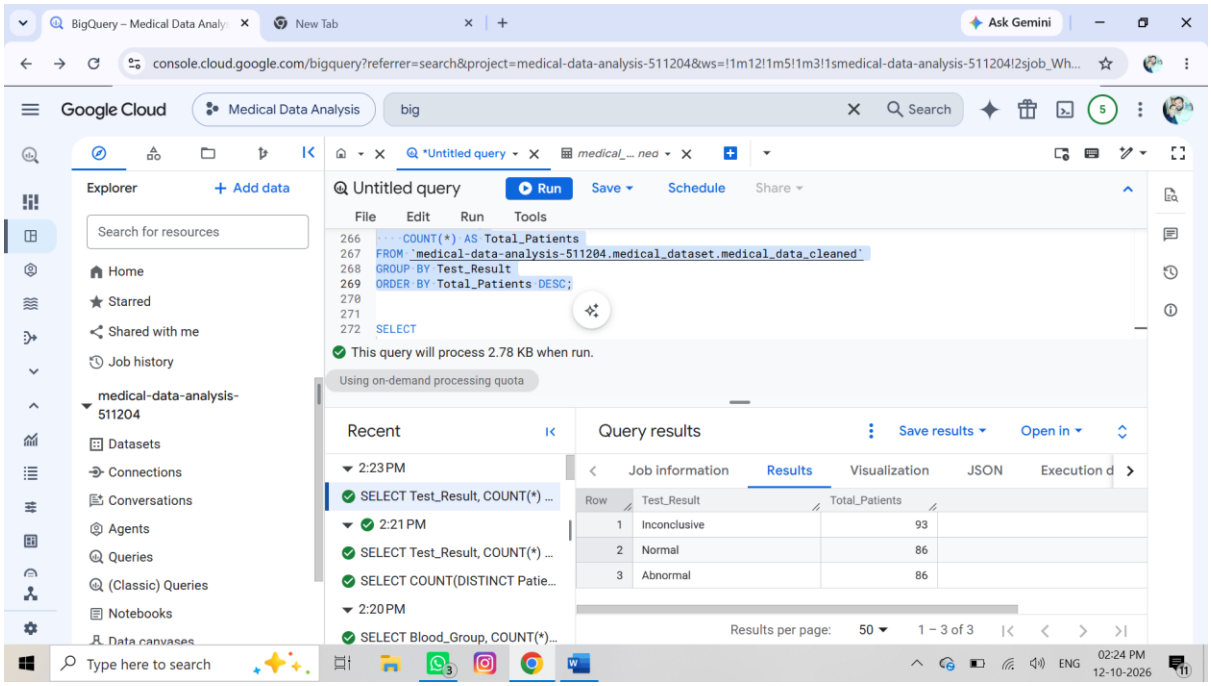
Test_Result,

COUNT(*) AS Total_Patients

FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`

GROUP BY Test_Result

ORDER BY Total_Patients DESC;



Final Business Insights

```
SELECT
   COUNT(DISTINCT Patient_ID) AS Total_Patients,

   ROUND(AVG(Age), 2) AS Average_Age,

   ROUND(AVG(Billing_Amount), 2)
      AS Average_Billing_Amount,

   ROUND(SUM(Billing_Amount), 2)
      AS Total_Billing_Amount,

   ROUND(AVG(Length_of_Stay_Days), 2)
      AS Average_Stay_Days,

   COUNT(DISTINCT Hospital) AS Total_Hospitals,

   COUNT(DISTINCT Medical_Condition)
```

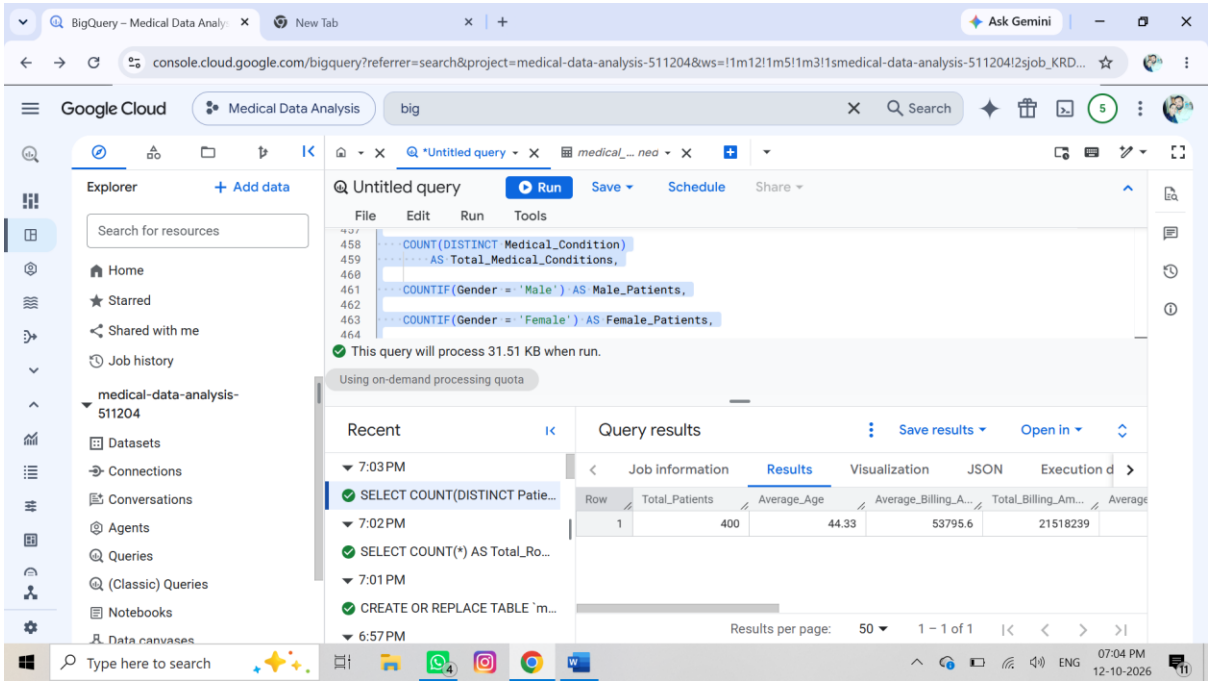
```
    AS Total_Medical_Conditions,

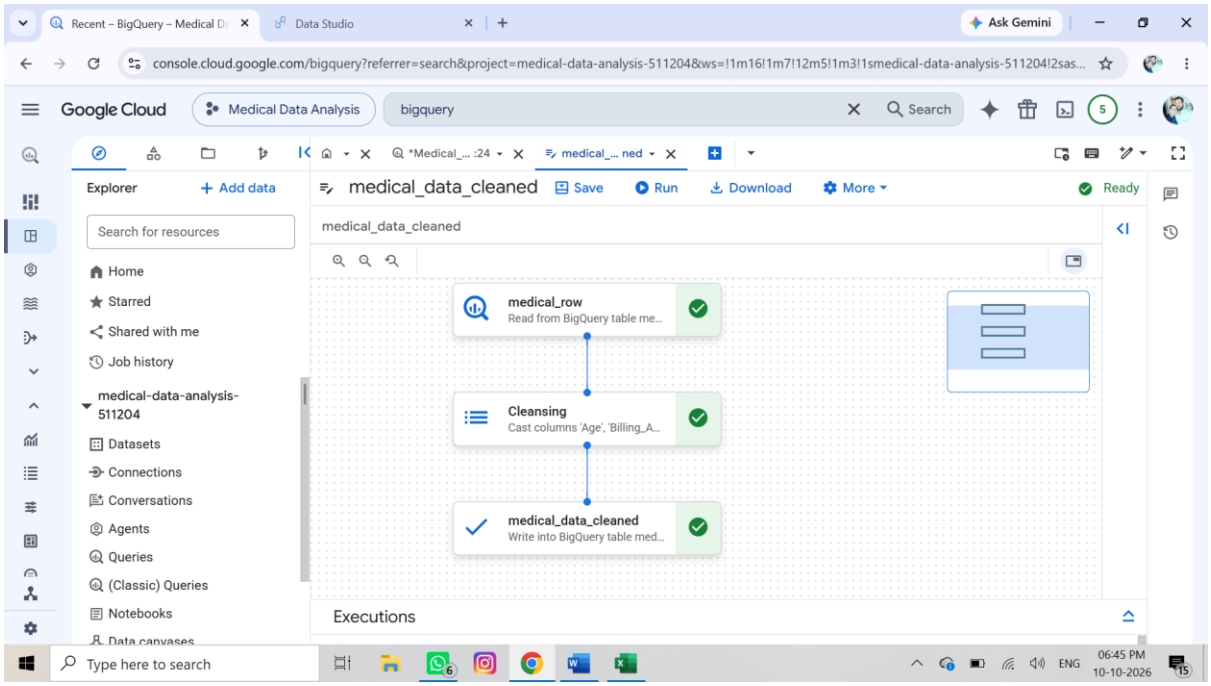
  COUNTIF(Gender = 'Male') AS Male_Patients,

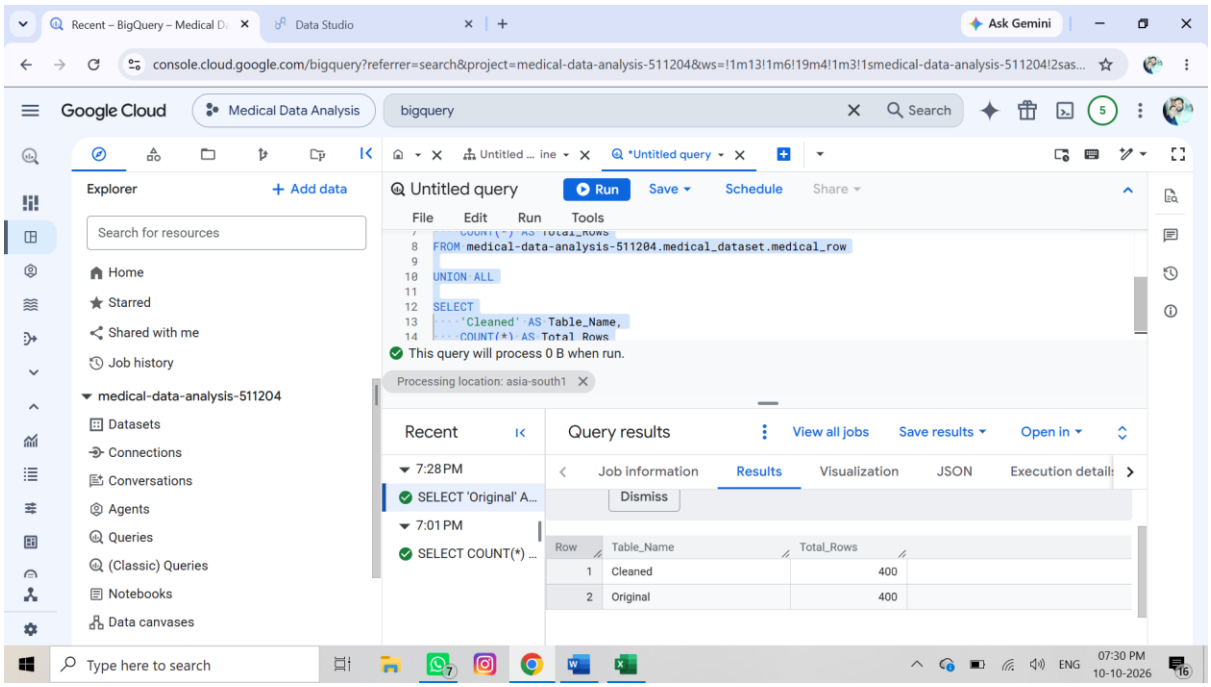
  COUNTIF(Gender = 'Female') AS Female_Patients,

  COUNTIF(Gender = 'Others') AS Other_Gender_Patients

FROM `medical-data-analysis-511204.medical_dataset.medical_data_cleaned`;
```



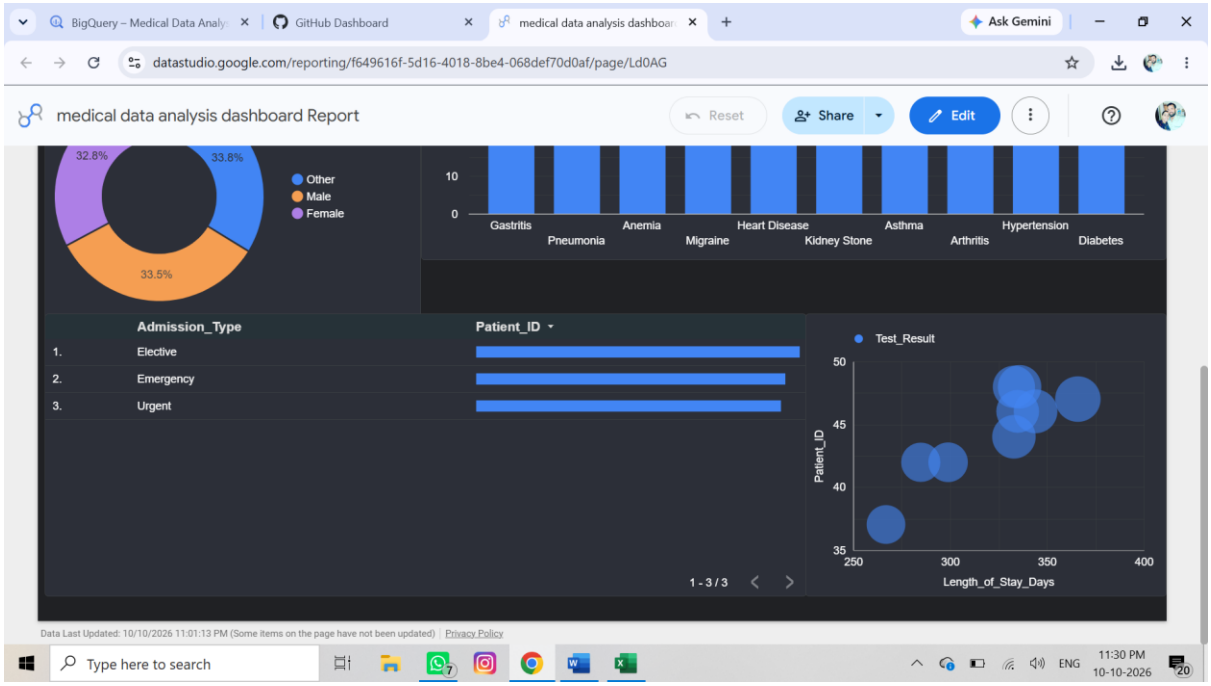


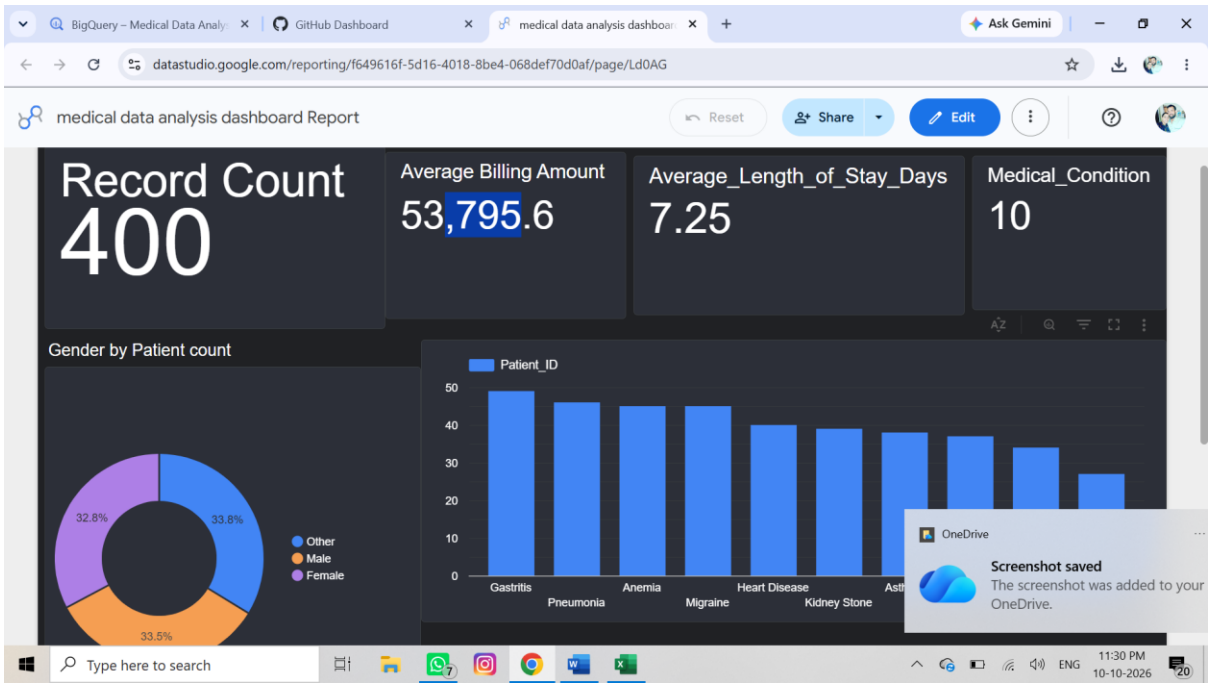


My Analysis

The medical dataset was analyzed using Google Cloud BigQuery and SQL to understand patient demographics, medical conditions, hospital admissions, and billing information. The analysis covered total patients, gender-wise distribution, age groups, medical condition-wise patient counts, admission types, hospital-wise patient counts, average billing amount, average length of stay, and blood group distribution. These analyses help identify patient trends, understand hospital service usage, and evaluate billing patterns.

Dashboard:using DATA STUDIO





Finding Insights – Medical Data Analysis

Total Patients: The dashboard displays a total patient count of 400, providing an overview of the patients included in the analysis.

Gender-wise Analysis: The dashboard shows three gender categories: Other (33.8%), Female (32.8%), and Male (33.5%). The distribution is relatively balanced across the categories.

Medical Condition Analysis: The dashboard compares patient records across medical conditions, including Gastritis, Pneumonia, Anemia, Migraine, Heart Disease, Kidney Stone, Asthma, Arthritis, Hypertension, and Diabetes.

Admission Type Analysis: The dashboard includes Elective, Emergency, and Urgent admission types, allowing comparison of different hospital admission categories.

Patient and Hospital Analysis: Patient-related charts help explore patient records and compare patterns across the available data.

Length of Stay Analysis: The dashboard includes a chart comparing Length of Stay (Days) with Patient ID, helping visualize the recorded stay-duration values.

Overall Insight: The dashboard brings patient demographics, medical conditions, admission categories, and length-of-stay information together to support healthcare data analysis and reporting.

Conclusion

This Medical Data Analysis project was completed using Google Cloud BigQuery, SQL, and Looker Studio. The analysis of 400 patient records provided insights into patient demographics, gender distribution, medical conditions, admission types, and length of hospital stay. An interactive dashboard was developed to visualize healthcare data and present key information clearly. This project demonstrates how data analytics can help identify healthcare trends, monitor hospital operations, and support data-driven decision-making. Overall, the project improved my practical knowledge of SQL queries, cloud-based data analysis, and dashboard development.